

NEURAL BELLMAN OPERATORS

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This paper develops Neural Bellman Operators for policy evaluation and feasible improvement in controlled economies. Centered continuation errors and own-policy comparisons connect numerical approximation to full-policy loss. A constructive nonlinear theorem covers continuous states and constrained actions under recursive risk, and an endpoint verifier certifies trained ReLU continuations over all states. A complementary quadratic construction links hidden-factor updates to economic error budgets. Residual enclosures make its mixed-precision implementation verifiable without exact matrix solves. A factorial study separates precision from caching and reports first-pass economic accuracy, failed checks and complete work. In a nonlinear investment economy, independent policy evaluation certifies gains from reoptimization after a change in investment costs. Conventional structural and spline methods remain faster in the reported comparisons. The controlled-diffusion framework and applications to recursive utility, endogenous preferences, temporal selves and games remain part of the same operator theory, with explicit application-specific conditions.

KEYWORDS: Dynamic programming, Neural approximation, Recursive utility, Endogenous preferences, Viscosity solutions, Dynamic games.

1. INTRODUCTION

Dynamic economic decisions depend on continuation values that are costly to recompute and change when future policies, technologies, or preferences change. A neural approximation can make repeated evaluation inexpensive, but neither a small fitting loss nor a successful current decision establishes that the resulting dynamic policy is accurate. The relevant questions are which continuation errors affect decisions, how those errors propagate through the policy's own future, and

what work is needed to construct and verify a neural continuation at the required economic accuracy.

This paper develops Neural Bellman Operators to answer these questions within a single policy-evaluation and feasible-improvement framework. The original object is a controlled economy with a policy-specific continuation, including controlled diffusions and recursive preferences. We retain the distinction between evaluating a fixed policy and optimizing against that evaluation. Current actions are varied over their actual feasible set; the future policy is held fixed while its continuation is evaluated. Boundary, utility-domain, and comparison conditions belong to the economic problem and are not replaced by a neural parameterization. Two distinct realizations now give constructive policy certificates: a recursive capital economy with Gaussian innovations and vector controls, and a nonlinear compact-state investment economy with constrained actions. The latter admits genuinely non-quadratic continuations and trained ReLU policies. Both are applications of the same operator framework.

1.0.0.1. Contribution and roadmap. Section 11 proves an all-adapted-policy comparison and a finite nonlinear construction with explicit off-grid and continuous-action allowances. It also supplies a breakpoint verifier for trained hidden units. Section 12 certifies the entire native matrix step and accounts for evaluation and verification work. Section 13 uses the complete factorial design and develops a nonlinear investment-price counterfactual. References below to the eight earlier exact services describe a historical construction audit; their joint adaptive-cached contrast does not identify a precision effect. The new factorial does not establish a stable adaptive timing advantage.

The first contribution identifies the decision-relevant error. An additive level shift in a continuation does not change an optimizer. The appropriate local error is therefore an oscillation across feasible actions, together with any optimization error and any discrepancy between the evaluated and implemented futures. This distinction matters for both theory and numerical diagnostics: a method may have a large level error but a small decision loss, or a small average fitting loss and an

inaccurate action. The centered bounds also explain why a certificate for a current action cannot simply be reused as a certificate for a changed future.

The second contribution gives a constructive route from trained hidden weights to full-policy accuracy. In the recursive capital economy, an independent coefficient comparison applies on all real states and against all real vector controls. The comparator is every adapted policy with finite recursive cost, not merely policies in the fitted class. Positive Gaussian moment margins control the propagation of quadratic and state-independent errors separately. A primitive envelope allocates policy error before the neural targets are supplied. Cold construction, a gradient-curvature criterion, and basin-certified warm construction then provide explicit routes to the required continuation accuracy. These statements concern an explicitly trainable square-activation continuation; every entry of its hidden factor may change. They are not fixed-feature regression guarantees applied to an unrelated jointly trained network.

The third contribution joins that construction to finite-precision reuse. A changed continuation may invalidate the old action while leaving the old factor in an admissible training basin. We derive a two-sided error budget that pays separately for factor fitting, own-policy evaluation, and the actor equation. It also checks the magnitude of the fitted coefficient, preventing an overshooting factor from invalidating the primitive envelope used to prove the policy bound. Within this budget, a precision-adaptive refresh rule permits rectangular factors and noncommuting starts. Its relative Gram enclosure contracts according to $\rho_{j+1} = \rho_j^2$ when the entire implemented step error receives the stated allowance. A fixed target inverse is constructed at most once per fit; subsequent applications and checks remain charged. The resulting theorem specifies both a finite update cap and a sum of stored fractional precisions. It does not presume that ordinary floating-point matrix operations have zero error.

These results supply a positive neural construction theorem, not a claim that neural methods must dominate every structural solution. We distinguish three statements throughout the paper. A policy certificate establishes the accuracy of a returned policy. A training theorem establishes that the declared neural construction can produce an admissible continuation under stated primitive and initialization

conditions. A complete-work comparison establishes which executed procedure was less costly at a common economic target. None of these statements alone establishes the other two. In particular, a Gaussian quadratic economy has a strong structure-exploiting competitor, and we retain it rather than make the comparison artificially favorable to learning.

The numerical evidence makes these distinctions observable. The earlier finite-law study establishes valid current-decision certificates and detects continuation changes, but its adaptive neural rule is more costly than fresh neural refitting in all eight reported dimension–volume cells; cached simulation is the least-cost fully certified procedure in those cells. Those outcomes and the complete original economic applications remain part of the publication. Subsequent continuous-support, vector-control studies execute full-policy comparisons, including a 72-service recursive-risk catalogue. The new frozen comparison isolates construction, precision, and target reuse on eight exactly specified services. All eight satisfy the same full-policy tolerance. Precision adaptation reduces storage precision and recorded complete service time relative to the two inherited neural constructions in both specified regimes. The structural procedure is faster still. These are deterministic construction comparisons, not estimates of a population probability of neural superiority.

The economic interpretation likewise depends on the target. A current withdrawal response conditional on a specified future policy differs from a counterfactual in which all future controls are reoptimized. We retain the former finite-law comparisons with their original interpretation and use the returned policy’s own continuation in every date of the latter recursive construction. The capital primitives are specified theoretical economies rather than empirical estimates. The separate applications to endogenous preferences, recursive utility, temporal selves, and games retain their own utility domains and deviation or equilibrium conditions. The capital certificate does not silently discharge those different obligations.

The article proceeds from the economic model and Neural Bellman Operator to decision errors, full-policy comparison, constructive training, and precision-adaptive reuse. The technical supplement supplies the proofs and supporting training criteria. The complete previous article, its supplement, and all economic applications are preserved as identified companions, with cross-document references. This orga-

nization permits the current argument to be read in economic and mathematical order without discarding earlier results or requiring the reader to reconstruct the development from branch history.

1.1. *Relation to the literature*

Neural Bellman Operators build on numerical dynamic programming and function approximation. Projection and related methods exploit economic structure to approximate a Bellman equation; see [Judd \(1998\)](#). Smooth neural networks can approximate derivatives under representation hypotheses ([Hornik et al., 1990](#)). These results distinguish representational capacity from the accuracy achieved by a finite architecture and a particular optimizer.

The differential critic is related to the neural PDE approach in [Sirignano and Spiliopoulos \(2018\)](#). Their DGM formulation includes differential and boundary conditions and treats HJB equations. The deep BSDE method of [Han et al. \(2018\)](#) also includes an HJB application. Our object combines a specified policy continuation, a feasible actor, and an economic error account, and asks when the continuation representation is useful across decision queries. An analytical maximizing action is useful when available; it is not a universal distinction between neural PDE methods. Recursive dependence on the value is a nonlinear zeroth-order term, whereas full nonlinearity of a PDE concerns its highest derivatives.

Recursive utility separates intertemporal substitution from risk aversion ([Epstein and Zin, 1989](#), [Duffie and Epstein, 1992](#)). Utility-process selection and transversality matter in its infinite-horizon formulation ([Herdegen et al., 2021](#)). Temporal selves instead require an equilibrium formulation ([Björk et al., 2016](#)). The stochastic approximation literature provides convergence results under explicit step-size, stability, and limiting-dynamics conditions ([Borkar, 2024](#)). We retain those distinctions when describing a finite neural computation.

Section 2 defines the economic object and Section [S.31](#) specializes it to the capital economy. Section [S.31.3](#) connects continuation error with economic gain. The computational design in Section [S.2](#) distinguishes continuation reuse, conditional payoff replication, and mechanism assessment. Section [15](#) states the other economic

applications. General verification results and their full proofs remain in the technical supplement, beginning with Section S.5; the application companion is a current part of the paper.

1.2. Continuation approximation and the contribution of NBO

A reusable continuation is not a new principle of dynamic programming. Postdecision-state approximation separates a controlled transition from the subsequent uncertainty and is a standard device in approximate dynamic programming (Powell, 2011). Regression can estimate a continuation relevant to an economic stopping decision, as in Longstaff and Schwartz (2001). Batch fitted value methods turn transition data into successive supervised regression problems (Ernst et al., 2005); approximate policy iteration likewise separates an evaluation error from an imperfect improvement step (Scherrer et al., 2015). These methods establish the appropriate comparison class for NBO. The relevant question is not whether continuation fitting is possible, but which fitted object, decision rule, and verification account are useful for the stated economic exercise.

The actor-critic distinction is also established. In Konda and Tsitsiklis (1999), a critic supplies information for a parameterized policy update under explicit stochastic-approximation conditions. NBO uses a scalar policy continuation and its derivative channels in a feasible economic improvement map. The scalar representation couples value and gradient predictions through one potential; a separately fitted vector field need not have this property. That structural distinction is testable but does not imply that a scalar neural network must outperform a costate surrogate, a direct actor, or a conventional scalar regression. The menu's vector baseline and the new quadratic, radial-basis, and shared-actor baselines therefore remain in the primary comparisons.

The reuse comparison is closely related to simulation optimization. A cached sample-average objective can be optimized at many current tasks. Reusing innovations is legitimate; reusing their payoff at a different postdecision state without reevaluation is not. The new experiment gives cached simulation and regression the same finite-law information. The regressions share their value and gradient

labels, and the shared actor receives the same task variables. Full-support enumeration is a particularly strong comparator when the law has only sixteen atoms. Its tractability is disclosed rather than suppressed to manufacture an advantage for approximation.

Several mathematical components are standard. Centering removes constants from action comparisons; the squared-error difference cancels common noise; shortest-path closure solves difference constraints; and a strong-concavity bound converts a first-order residual into an objective gap. For stochastic stopping, time-uniform confidence sequences provide a general route to valid repeated assessment (Howard et al., 2021). We do not claim any of these principles as a new theorem of dynamic programming, graph theory, or probability. The contribution developed here is their explicit economic linkage: which continuation is reusable, how approximation and changes in future primitives enter decision loss, which computed action is certified, and which work is actually paid before that certificate is returned. The deterministic finite-law experiment uses no confidence sequence and spends no additional probability. Its scalar-interval guarantee and the earlier statistical menu guarantee are different applications of that linkage.

2. THE CONTROLLED ECONOMY

Let W be an m -dimensional Brownian motion on a filtered probability space satisfying the usual conditions. The state $S = (u, X)$ contains internal and external state variables. An action $a = (\theta, c, \pi, \dots)$ combines internal adjustment and external choices. The controlled state evolves as

$$dS_s = b(s, S_s, a_s) ds + \sigma(s, S_s, a_s) dW_s, \quad a_s \in \mathcal{A}(s, S_s). \quad (1)$$

The compact feasible correspondence $\mathcal{A}$ is part of the model. Actions are progressively measurable and must generate a well-defined state process satisfying the declared state restrictions. State viability does not follow from bounded actor outputs alone.

We work initially on a finite horizon T . A computational domain D is either invariant, equipped with a specified reflection mechanism, or stopped on exit. In the stopped case let $\tau = T \wedge \inf\{s \geq t : S_s \notin D\}$. The payoff $g(\tau, S_\tau)$ is specified on

the entire parabolic stopping boundary, including the terminal face. For a fixed admissible policy a , utility is the policy-specific solution

$$Y_s^a = g(\tau, S_\tau) + \int_s^\tau f(v, S_v, a_v, Y_v^a) dv - \int_s^\tau Z_v^a dW_v. \quad (2)$$

Adjustment costs are included in f . For example, time-additive utility has $f(s, S, a, y) = r_0(s, S, a) - \rho y$, where r_0 includes all current resource and adjustment costs. Define $V^a(t, x) = Y_t^{a, t, x}$ and $V^*(t, x) = \sup_a V^a(t, x)$. The value inside a candidate policy's recursive objective is its own utility process, not the unknown optimum inserted into every candidate objective.

ASSUMPTION 1—Finite-horizon comparison class: *The controlled state problem is well posed for the admissible policies used below. The coefficients are continuous, locally Lipschitz in the state uniformly over actions, and bounded on the compact stopped domain. The payoff is continuous and bounded. On an invariant utility interval containing all policy values, f is continuous in its arguments and uniformly Lipschitz in utility with constant $L \geq 0$. The admissible class is stable under concatenation and has the measurable-selection properties needed for dynamic programming. The resulting HJB problem and the fixed-policy problems obey comparison in the bounded continuous class with the specified boundary conditions.*

The comparison requirement identifies both a function class and a boundary problem. It is not asserted merely because the primitive coefficients are continuous. On a bounded, regular Dirichlet domain, a boundary barrier is one way to verify attainment of the stopping payoff. Reflection requires the corresponding Neumann or oblique derivative condition and a well-posed reflected state process. Unbounded-state applications require growth bounds and localization. These conditions are checked separately from a neural residual.

For a smooth function v , write

$$\mathcal{L}^a v = b(t, x, a)^\top Dv + \frac{1}{2} \text{tr}\{\sigma(t, x, a)\sigma(t, x, a)^\top D^2 v\}, \quad (3)$$

$$\mathcal{H}^a[v] = f(t, x, a, v) + \mathcal{L}^a v, \quad F[v] = -v_t - \sup_{a \in \mathcal{A}(t, x)} \mathcal{H}^a[v]. \quad (4)$$

The value solves $F[V^*] = 0$ with the specified boundary data, in the viscosity sense where classical derivatives do not exist. The diffusion covariance is the positive semidefinite matrix $\sigma\sigma^\top$. No claim that the product of two arbitrary positive semidefinite matrices is symmetric positive semidefinite is needed.

A continuous martingale representation can also be used when its quadratic variation is Markov in an augmented state. If $d\langle M \rangle_s = Q(s, S_s, a_s)ds$, the covariance in (3) becomes $\sigma Q \sigma^\top$. A merely predictable process Q_s is not, without further state augmentation, a deterministic coefficient of a Markov HJB equation. Quadratic variation by itself also does not imply conditionally Gaussian increments. The implementation in this paper uses the Brownian model (1).

2.1. Infinite horizon and economic normalization

For a stationary problem, the boundary-value equation is $\sup_a \{f(x, a, V) + \mathcal{L}^a V\} = 0$. We use an infinite-horizon result only after specifying admissibility, integrability, and the selected utility process. In the time-additive case, a sufficient verification requirement is that the localized Itô identities are uniformly integrable and $\mathbb{E}[e^{-\rho T} |V(S_T)|] \rightarrow 0$ for the policies being compared. Recursive problems require an appropriate utility comparison and transversality condition instead of simply borrowing the additive one.

Cardinal transformations also require care. Multiplying a fixed utility function by a positive constant does not change its maximizer. Adding a function of an *endogenously controlled preference state* generally does. Consumption units, preference normalization, and terminal utility are therefore primitive economic choices in Section A.1.3.

3. NEURAL BELLMAN OPERATORS

The actor $a_\phi(t, x)$ and critic $v_\zeta(t, x)$ represent policies and continuation values. An internal actor and an external actor can be separate parameter blocks; their joint action remains the input to the same economic generator. A smooth activation is useful for exact differentiation, while a nonsmooth neural representation can be used with the positive-weight Bellman formulation below.

3.1. Feasibility and boundary data

For a box $[a, \bar{a}]$, a sigmoid map produces interior feasible actions but does not attain its endpoints at finite weights. Binding constraints can instead be represented by a projection onto the closed interval, by an explicit boundary candidate, or by a finite action set including the endpoints. The constrained portfolio calculation uses a projection and can attain a zero risky share exactly. On a simplex, a probability map supplies a feasible interior point and boundary faces must again be considered when certifying the optimum. General constraints require a feasible parameterization, a certified projection, or a constrained action solver. A finite penalty is not an exact feasibility certificate.

On a finite horizon, an elementary hard-terminal architecture is

$$v_{\xi}(t, x) = G(x) + (T - t)N_{\xi}(t, x). \quad (5)$$

This enforces the terminal face, but not automatically a lateral boundary. A compatible lifting G_B and a factor d_B vanishing on all Dirichlet faces give $v_{\xi} = G_B + d_B N_{\xi}$. Alternatively, boundary residuals are trained separately and retained explicitly in the error bound. A reflected preference state requires its own derivative condition; a terminal architecture does not enforce it. For recursive utility, a domain-preserving transformation must additionally keep the continuation value in the domain of the aggregator.

3.2. The block-specific computational graph

Let ν be a specified distribution of state–time points and let sg denote stop-gradient. A critic step minimizes

$$L_C(\xi; \bar{\phi}) = \mathbb{E}_{\nu} \left[-\partial_t v_{\xi} - \mathcal{H}^{\text{sg}(a_{\bar{\phi}})}[v_{\xi}] \right]^2 + L_B(\xi), \quad (6)$$

where L_B is absent only for boundary conditions enforced exactly. The policy, sampling locations, and any rival actions are detached. The value, time derivative, state gradient, and Hessian remain differentiable with respect to the critic parameters.

An actor step minimizes

$$L_A(\phi; \bar{\xi}) = -\mathbb{E}_v \mathcal{H}^{a_\phi}[\text{sg}(v_{\bar{\xi}}), \text{sg}(Dv_{\bar{\xi}}), \text{sg}(D^2v_{\bar{\xi}})]. \quad (7)$$

All critic derivative channels are fixed in this step. There is no critic gradient from L_A and no actor gradient from L_C . A printed sum $L_A + \omega L_C$ may be a reporting device, but it is not jointly differentiated. The implementation exposes separate functions and regression tests for these derivative paths.

The distinction is substantive: joint differentiation can move the critic away from the policy-evaluation root. Appendix S.7 gives a counterexample, the recursive lifetime policy derivative, and the conditioning requirements. The local Hamiltonian objective is not identified with a lifetime policy gradient under arbitrary state sampling. The evaluated reference policy may remain fixed while a separate candidate actor is improved at successive work levels. Updating that reference defines a new evaluation target and is recorded explicitly.

3.3. Rollout and costate evaluation

For a frozen policy $a^{(j)}$, a positive-time Euler rollout from (t, y) supplies a sample return $\widehat{V}^{(j)}(t, y)$ and its pathwise derivative $\widehat{p}^{(j)}(t, y)$. Differentiation includes the state dependence of subsequent policy actions. Freezing the policy means freezing its parameters, not deleting that derivative. The targets evaluate the current policy, not the optimal policy, and contain no maximizing action labels. For fixed finite weights, bounded actions and smooth bounded production justify pathwise differentiation of this finite rollout away from the harmless clipping kinks. These are derivatives of the discretized evaluation problem, not asserted unbiased derivatives of its continuous-time limit.

The critic minimizes

$$\widehat{\mathbb{E}} \left[0.05 \{v_\theta - \widehat{V}^{(j)}\}^2 + \|d(D_y v_\theta - \widehat{p}^{(j)})\|_d^2 \right]. \quad (8)$$

The factor d puts normalized marginal values on an order-one scale. The rollout implementation uses two hidden tanh layers and a hard terminal component.

Holding its new derivative fixed, the actor then maximizes

$$\hat{\mathbb{E}} \left[\overline{\log a_\phi} - \frac{\zeta}{2} \bar{a}_\phi^2 - a_\phi \cdot \text{sg}(D_y v_\theta) \right]. \quad (9)$$

Terms independent of actions have been omitted. Consumption is represented directly by $m_0(t) + \epsilon \tanh N_\phi(t, y)$; there is no online Hamiltonian search in its deployment. Value evaluation, costate evaluation, and feasible improvement are distinct operations, with their own stored losses and work counts. The number of critic and actor steps per rollout batch is a declared design parameter. Failure of either numerical block is recorded, not interpreted as successful policy improvement.

The stochastic gradient targets make the economically relevant marginal continuation values visible to the evaluation loss. This is not a new Bellman principle, nor an assumption that derivative regression invariably dominates differential residual fitting. It is a concrete implementation of the NBO evaluation map. The preserved differential implementation and its two independent trace banks continue to test a different evaluation procedure. Both feed the same economic distinction between policy evaluation and feasible improvement.

3.4. The implemented map and its verification output

Fix the economy, boundary realization, network classes, optimization budget, and verification procedure, denoted collectively by ϑ . Let $\mathcal{P}$ be the admissible feedback policies, Ξ the critic parameter space, and Ω the sampling streams. The implemented NBO is the numerical map

$$\mathcal{N}_\vartheta : \mathcal{P} \times \Xi \times \Omega \longrightarrow \mathcal{V}_\vartheta \times \mathcal{P}_\vartheta \times \mathcal{C}_\vartheta, \quad (a, \xi, \omega) \longmapsto (v, \tilde{a}, C). \quad (10)$$

The first component is approximate policy evaluation; the second is feasible improvement; the third records verification of the resulting pair. A record identifies the model, parameters, domain, boundary faces, evaluation error, maximization error, coverage, discretization errors, and unchecked assumptions. An unavailable bound is not zero. Failure to obtain a useful certificate is a possible output of the map. With its sampling stream fixed, the implementation is deterministic. We do

not assert that a network learns an unrestricted Bellman operator on a function space.

The evaluation–improvement structure is an instance of approximate policy iteration, including its partial-evaluation variants (Scherrer et al., 2015). Neural representation does not introduce a new principle of optimality, and monotonicity belongs to the underlying positive-weight scheme, not to gradient descent. The NBO specification joins the block-specific derivative graph to boundary and utility-domain restrictions and to a common economic error account. Sections S.15.1 and A.1.3.4 implement the differential and positive-weight versions of this map, respectively, with independent verification of their outputs.

4. ECONOMIC DECISIONS AND ACCURACY TARGETS

A task θ and observed state s determine a feasible action set $A_\theta(s)$ and current reward $r_\theta(s, a)$. The postdecision state is $T_\theta(s, a)$. Write

$$Q_\theta(s, a) = r_\theta(s, a) + S_\theta\{T_\theta(s, a)\}, \quad L_\theta(s, a) = \sup_{b \in A_\theta(s)} Q_\theta(s, b) - Q_\theta(s, a). \quad (11)$$

The continuation S_θ includes future rewards under a specified future policy, conditional on the postdecision information. Current rewards are excluded. A change in the current charge alone need not change S_θ ; a change in the future policy generally does. For an observed catalogue $\{(s_i, \theta_i)\}_{i=1}^q$, our new primary target is $q^{-1} \sum_i L_{\theta_i}(s_i, a_i) \leq \varepsilon$.

The word “regret” below always refers to the displayed comparison class. The current scalar experiment takes $s \in \mathbb{R}^d$ but restricts the current withdrawal to a uniform scalar action. Neither a large state dimension nor full enumeration of its finite innovation law turns this target into an optimum over vector actions or all adapted future controls.

Algorithm 1 Neural Bellman evaluation, feasible improvement, and economic assessment

Require: Economic primitives and information; feasible actions and utility domain; boundary data; evaluation variant; training, validation and stopping rules; independent final assessment.

- 1: Declare the feasible reference policy to be evaluated. Initialize a feasible candidate and a boundary-compatible continuation representation.
 - 2: **for** each permitted evaluation-improvement iteration **do**
 - 3: Freeze the declared reference policy, rivals and evaluation target. Fit only the critic by the declared differential, positive-weight, or rollout rule. The reference may remain fixed across work levels.
 - 4: Freeze the critic and its value/derivative channels. Update only the feasible actor, or apply the declared feasible action solver.
 - 5: Record all evaluation and improvement work, corrections, unsuccessful checks, and the resulting deployment rule.
 - 6: Apply the predetermined validation and stopping rule, using no final-test outcomes.
 - 7: **end for**
 - 8: Freeze the selected policy and continuation object before independent final evaluation.
 - 9: Compute the specified economic endpoints and direct contrasts, with their approximation, implementation and statistical accounts.
 - 10: Return the fitted objects, their implementation and domain, and the complete accuracy and work record.
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5. FROM A LEARNED CONTINUATION TO ECONOMIC ACCURACY

5.1. *Centered approximation error and the selected decision*

Fix an anchor continuation S_0 and its approximation $\hat{S}_0$. For each task, let $e = \hat{S}_0 - S_0$ and $\Delta_\theta = S_\theta - S_0$. The implemented action $\hat{a}$ satisfies

$$r_\theta(s, \hat{a}) + \hat{S}_0(T_\theta(s, \hat{a})) \geq \sup_{a \in A_\theta(s)} \{r_\theta(s, a) + \hat{S}_0(T_\theta(s, a))\} - \delta_\theta(s). \quad (12)$$

TABLE I
ECONOMIC TARGETS AND THEIR VERIFICATION DOMAINS

Target	Comparison object	Interpretation in this paper
Continuation risk	Exact conditional continuation at specified arguments	Prediction diagnostic; neither policy welfare nor control regret.
Finite-menu loss	Best action in one exogenous finite action menu	Decision diagnostic at common tasks; not a continuous-action bound.
Catalogue regret	Best of sixteen historical method–budget candidates	Earlier simultaneous payoff intervals and their exact closure.
Scalar-interval regret	True maximum over $[\cdot, \bar{a}_i]$ in the stored finite economy	New primary accuracy certificate and realized stopping target.
Diffusion control regret	Full declared adapted-control class in continuous time	Separate boundary, implementation, diffusion and information accounts.
Utility or game criterion	Policy utility, spike deviation or unilateral best response	Application-specific admissibility and verification conditions.

Here $\delta_\theta(s) \geq 0$ is an actual optimization allowance, not a training-loss tolerance. For a function on the feasible actions, write $\text{osc}_A f = \sup_A f - \inf_A f$.

PROPOSITION 1—Continuation error, transport, and decision loss: *Suppose the suprema in (11) and (12) are finite. Then*

$$L_\theta(s, \hat{a}) \leq \delta_\theta(s) + \text{osc}_{A_\theta(s)}(e \circ T_\theta) + \text{osc}_{A_\theta(s)}(\Delta_\theta \circ T_\theta). \quad (13)$$

For a finite action set $\{a_1, \dots, a_m\}$ with $m \geq 2$ and fixed positive weights π_j summing to one, define

$$R_\pi(s) = \sum_j \pi_j (e(T_\theta(s, a_j)) - \bar{e}_\pi(s))^2,$$

$$\bar{e}_\pi(s) = \sum_j \pi_j e(T_\theta(s, a_j)),$$

$$C_\pi = \max_{i \neq j} (\pi_i^{-1} + \pi_j^{-1}).$$

If $|\Delta_\theta| \leq b_\theta$ on the queried postdecision set, then

$$L_\theta(s, \hat{a}) \leq \delta_\theta(s) + \sqrt{C_\pi R_\pi(s)} + 2b_\theta. \quad (14)$$

Both inequalities hold for each fitted object and each realized cache. Expectations over training or cache randomness may subsequently be taken.

The proof is in Section S.1. An additive continuation error common to all feasible actions has no effect on the decision. This is why an absolute squared prediction error may be large even when decisions are accurate. Conversely, the risk at only the NBO/reference action pair cannot control the losses of another method choosing elsewhere. Our new diagnostic uses a common exogenous action set and, separately, the arguments selected by each predictor. Inequality (14) is a bound for the common finite set; it is not applied to the entire scalar interval using just three sampled actions.

If the true finite-menu best action has gap $g > 0$ from every competitor, then a bound strictly smaller than g implies that the implemented action is best. Without such a margin, arbitrarily small centered prediction error can change the identity of the selected action. Neither statement orders two methods' realized payoffs from their risks alone.

5.2. A nontrivial reuse region when the future changes

Let $X_{k+1}^\theta = F_{\theta,k}(X_k^\theta, Z_k)$ describe the future under the policy being evaluated. A common-innovation coupling need not make the transition maps identical. On an invariant state domain, assume

$$\begin{aligned} \|F_{0,k}(x, z) - F_{0,k}(x', z)\| &\leq L_k \|x - x'\|, & \|F_{\theta,k}(x, z) - F_{0,k}(x, z)\| &\leq \epsilon_{\theta,k}, \\ |g_{0,k}(x) - g_{0,k}(x')| &\leq H_k \|x - x'\|, & |g_{\theta,k}(x) - g_{0,k}(x)| &\leq \eta_{\theta,k}. \end{aligned} \quad (15)$$

These bounds concern future rewards and transitions, not just the current task. Include the terminal payoff as $g_{\theta,K}$ and use nonnegative reward weights ω_k . The initial postdecision state is shared.

THEOREM 2—Continuation transport and refresh: Set $d_0 = 0$ and $d_{k+1} = L_k d_k + \epsilon_{\theta,k}$. Under (15), uniformly on the common postdecision domain,

$$|S_\theta - S_0| \leq b_\theta := \sum_{k=0}^K \omega_k (\eta_{\theta,k} + H_k d_k). \quad (16)$$

Thus the anchor may be reused at a task whenever the right side of (13), or the applicable finite-menu bound (14), is at most the desired economic tolerance.

More generally, suppose a recursive evaluation on a common invariant utility domain has maps $V_{\theta,k} = \mathcal{G}_{\theta,k}(V_{\theta,k+1})$. If these maps are sup-norm Lipschitz with constants β_k on that domain, $\|\mathcal{G}_{\theta,k}(v) - \mathcal{G}_{0,k}(v)\|_\infty \leq \nu_{\theta,k}$, and $\|V_{\theta,K} - V_{0,K}\|_\infty \leq b_K$, then

$$\|V_{\theta,0} - V_{0,0}\|_\infty \leq \sum_{k=0}^{K-1} \left(\prod_{j=0}^{k-1} \beta_j \right) \nu_{\theta,k} + \left(\prod_{j=0}^{K-1} \beta_j \right) b_K. \quad (17)$$

Empty products equal one.

The theorem supplies a class of tasks with genuinely different futures. For example, uniformly bounded transition and reward perturbations of an anchor economy satisfy it on any verified invariant compact domain with bounded Lipschitz constants. A single learned continuation then supports a whole neighborhood of counterfactuals until its economic error allowance is spent. The neighborhood is conditional on the displayed stability bounds; it is not a distribution-free assertion about all future policies. When future rewards, constraints, or opponents' policies violate those bounds, the evaluated continuation must be recomputed or its transport allowance reestablished. For recursive utility the common utility domain is essential: a local Lipschitz constant cannot be extended across a singular utility boundary. For games, this comparison evaluates a specified deviation problem; it does not replace the fixed-point or best-response conditions in the Economic Applications companion.

5.3. An explicit amortization class

The preceding inequalities give a sufficient class in which a shared scalar continuation supports a cheaper declared accuracy procedure, not just a post hoc

interpretation of successful cells. Fix a finite exogenous query family with K distinct postdecision arguments and a common continuation $S(x) = \phi(x)^\top \beta$. The feature map $\phi : \mathcal{X} \rightarrow \mathbb{R}^p$ may be a frozen neural dictionary or a conventional basis. Its construction cost is charged, and it is fixed before the fitting noises below. At design points x_i , observe $Y_i = S(x_i) + \xi_i$, where the noises are conditionally independent, mean zero, and σ -sub-Gaussian. Let Φ have rows $\phi(x_i)^\top$ and assume

$$G_n = \Phi^\top \Phi \text{ is invertible, } \max_{x \text{ queried}} \phi(x)^\top G_n^{-1} \phi(x) \leq \ell/n.$$

For example, $G_n/n \succeq cI$ and $\|\phi(x)\|^2 \leq B_\phi^2$ imply $\ell = B_\phi^2/c$. These are checkable design conditions, not a claim about an arbitrary trained representation.

COROLLARY 3—A sufficient learned-continuation work advantage: Fix $\varepsilon > \delta \geq 0$ and $0 < \alpha < 1$. Least-squares continuation fitting and δ -accurate finite-menu maximization have loss at most ε on every query with probability at least $1 - \alpha/2$ if

$$n \geq \frac{8\sigma^2\ell}{(\varepsilon - \delta)^2} \log \frac{4K}{\alpha}, \quad (18)$$

provided the displayed full-rank and leverage conditions hold. Direct sample-average predictions of the same continuation have the same guarantee when they use at least $t = \lceil 8\sigma^2 \log(4K/\alpha) / (\varepsilon - \delta)^2 \rceil$ observations per distinct argument, under the analogous scalar noise bound. Common innovations across arguments are allowed.

Suppose the complete declared learned procedure costs $F(n) + Kc_N + C(q)$ and the declared direct procedure costs $C_0(t) + Ktc_R + C(q)$, with all fitting, caching, selection and verification included. Both procedures meet the target jointly with probability at least $1 - \alpha$, and the learned procedure is strictly cheaper whenever

$$K(tc_R - c_N) > F(n) - C_0(t). \quad (19)$$

Repeated identical arguments are counted once in K . If verification or selection costs differ, their actual difference enters the right side.

The class contains nonlinear continuations represented by finite smooth neural features, not only linear economic models. The leverage bound keeps fitting

accuracy uniform as queries are reused. A bounded representation bias or trans-
ported future adds its explicit allowance to the decision bound and reduces the
available margin in (18). The result compares two fully specified sufficient-accuracy
procedures; it is not a lower bound on the best possible simulation method. A Raw
procedure may itself fit a surrogate, and then belongs to the learned comparison
class. In particular, this corollary neither certifies the assumptions for the experi-
ment’s jointly trained network nor asserts that its conventional surrogates must be
slower. It provides an analytical reuse class that the numerical results need not be
asked to establish by extrapolation.

5.4. Actual stopping and the amortization threshold

Consider a feasible scalar action in $[\ell, u]$ and a twice differentiable true objective
with $Q'' \leq -\kappa < 0$. Define the directional residual

$$r(a) = \begin{cases} \max\{Q'(a), 0\}, & a = \ell, \\ \max\{-Q'(a), 0\}, & a = u, \\ |Q'(a)|, & \ell < a < u. \end{cases}$$

An interval enclosure of $Q'(a)$ and a lower bound on κ give a computable upper
bound $U(a)$ in the following statement.

PROPOSITION 4—Certified stopping and complete work: *For every feasible scalar
action,*

$$\sup_{b \in [\ell, u]} Q(b) - Q(a) \leq \frac{r(a)^2}{2\kappa}. \quad (20)$$

*For a finite sequence of candidate-fitting stages, let a service stop at the first stage whose
mean outward bound is at most ϵ . If each bound is a deterministic enclosure for the true
task objectives, the returned service satisfies the target regardless of the preceding failed
checks. If checks instead have conditional error allocations α_j given the past and $\sum_j \alpha_j \leq \alpha$,
the analogous conclusion holds with probability at least $1 - \alpha$.*

*The realized cost of a service stopping at stage J is the sum of all initialization, fitting or
cache construction, querying, verification, and durable-record costs through J . In an exact*

fixed-plus-linear cost model $W_m(q) = F_m + qc_m$, where c_m includes marginal verification and all attempted stages, two methods that both meet the same target satisfy

$$W_N(q) < W_R(q) \iff q(c_R - c_N) > F_N - F_R. \quad (21)$$

When $c_R > c_N$, this gives the strict break-even threshold. If only cost intervals are available, a guaranteed ranking requires an upper bound for one method below a lower bound for the other, not two upper bounds.

The cost model is an interpretation, not an imposed shape for the measured frontier. Fitting stages may change with query volume, and verification can dominate marginal prediction work. We therefore execute every volume rather than extrapolate a line from a single 384-query comparison. The new experiment runs each method's frozen stage sequence; it does not search globally for a winner and then claim to have paid only that winner's bill. All comparison services and shared startup are reported separately from the service that an economist would deploy after choosing a method.

6. AN ALL-STATE POLICY CERTIFICATE

The sup-norm account above is natural on bounded value domains. In a capital model with unbounded shocks, a uniform bound on an absolute local advantage can instead be infinite. A cost-weighted account resolves this difficulty without restricting verification to states visited by the fitted policy. We state it for additive evaluation. The recursive-utility and strategic models retain the domain and deviation conditions in Section S.2 and the Economic Applications companion.

Use a cost-minimization convention in this section: a reward is the negative of the corresponding cost. Let $\ell_t(x, a) \geq 0$, $h(x) \geq 0$, and $0 < \beta \leq 1$. For an admissible policy π , define

$$U_t(x) = J_t^\pi(x) = \mathbb{E}_x^\pi \left[\sum_{s=t}^{T-1} \beta^{s-t} \ell_s(X_s, A_s) + \beta^{T-t} h(X_T) \right]. \quad (22)$$

The comparison class includes every adapted policy with finite cost. Assume that the conditional expectations below exist for these policies. The candidate's local

improvement opportunity is

$$g_t(x) = U_t(x) - \inf_a \{ \ell_t(x, a) + \beta \mathbb{E}[U_{t+1}(X_{t+1}) \mid x, a] \}. \quad (23)$$

In particular, g_t uses the candidate's actual future, not an optimal continuation or a fitted proxy for it.

THEOREM 5—Coercive policy certificate: *Suppose that $U_T = h$ and, for one finite $\eta \geq 0$,*

$$0 \leq g_t(x) \leq \eta \ell_t(x, a) \quad \text{for every date, state, and admissible action.} \quad (24)$$

Then, with J_t^ the infimum over all adapted finite-cost policies,*

$$0 \leq U_0(x) - J_0^*(x) \leq \frac{\eta}{1 + \eta} U_0(x). \quad (25)$$

The statement is deterministic when its premises are deterministic. A sampled certificate must establish all its premises on a common event.

PROOF: For any admissible action, the definition of g_t and (24) imply $U_t(x) \leq (1 + \eta) \ell_t(x, a) + \beta \mathbb{E}[U_{t+1} \mid x, a]$. Apply this inequality along an arbitrary adapted comparison policy and iterate conditional expectations. At the terminal date,

$$U_0(x) \leq (1 + \eta) \mathbb{E} \sum_{t < T} \beta^t \ell_t + \beta^T \mathbb{E} h \leq (1 + \eta) J_0(x),$$

where the last step uses $h \geq 0$. Taking the infimum over comparison policies and rearranging proves the assertion. The lower bound follows because the candidate is admissible. *Q.E.D.*

The argument is a Bellman-residual comparison, not a new principle of optimality. Its useful feature is that a bound proportional to a coercive stage cost can hold on an unbounded domain even when an absolute sup-norm bound cannot. It is not enough to check (24) on a finite catalogue. Nonnegative costs and the comparison-policy integrability requirements are substantive premises.

6.1. A multisector capital-adjustment economy

Let $x \in \mathbb{R}^d$ denote capital stocks relative to a reference path and $u \in \mathbb{R}^d$ their investment adjustments. There are T reoptimized dates:

$$X_{t+1} = A_t X_t + B_t u_t + \tilde{\xi}_{t+1}, \quad \tilde{\xi}_{t+1} \sim N(0, \Sigma), \quad (26)$$

$$\ell_t(x, u) = x' Q_t x + u' R_t u, \quad h(x) = x' Q_T x. \quad (27)$$

Shocks are independent across dates and correlated across sectors. Q_t and R_t are positive definite; Q_T is positive semidefinite. The problem is posed in discrete time, on the full state and action spaces. It is a quadratic tracking economy, not a calibrated quantitative description of a particular industry and not a discretization claim about the original diffusion model. Its continuous innovation support and vector controls make it a direct test of the policy certificate, rather than a larger scalar catalogue.

For a stored linear policy $u_t = -K_t x$, its *own* value is $U_t(x) = x' P_t x + c_t$, where, writing $F_t = A_t - B_t K_t$,

$$\begin{aligned} P_T &= Q_T, & c_T &= 0, \\ P_t &= Q_t + K_t' R_t K_t + \beta F_t' P_{t+1} F_t, & c_t &= \beta \{c_{t+1} + \text{tr}(P_{t+1} \Sigma)\}. \end{aligned} \quad (28)$$

This is evaluation of a supplied policy. It contains no optimization over future gains. Define

$$H_t = R_t + \beta B_t' P_{t+1} B_t, \quad D_t = H_t K_t - \beta B_t' P_{t+1} A_t. \quad (29)$$

Completing the square in the full vector action gives the exact identity $g_t(x) = x' D_t' H_t^{-1} D_t x$. In particular, an action mesh is unnecessary.

COROLLARY 6—Uniform capital-policy bound: Let $q_t \leq \lambda_{\min}(Q_t)$ and $r_t \leq \lambda_{\min}(R_t)$ be strictly positive lower bounds. Set

$$\eta = \max_{t < T} \frac{\|D_t\|_F^2}{q_t r_t}. \quad (30)$$

For every state, the stored policy satisfies (25). Consequently, on $\|x\|_2^2 \leq b$, its policy loss is at most

$$\varepsilon_{\text{pol}} = \frac{\eta}{1 + \eta} \{b\|P_0\|_\infty + c_0\}. \quad (31)$$

The comparison is with all adapted finite-cost policies and all real vector actions, not only linear policies or the class used to construct the critic.

PROOF: $H_t \succeq R_t \succeq r_t I$, so $g_t(x) \leq \|D_t\|_F^2 \|x\|_2^2 / r_t \leq \eta x' Q_t x \leq \eta \ell_t(x, u)$ for every u . Theorem 5 applies. The true P_0 is symmetric positive semidefinite; hence $x' P_0 x \leq \|x\|_2^2 \|P_0\|_\infty$. Finite-cost comparison policies have finite second moments at every relevant date by coercivity, which supplies the required integrability. *Q.E.D.*

The coefficient matrices, rather than sampled state outcomes, establish coverage. Thus the class-approximation allowance is zero: the verifier already compares with the unrestricted adapted class. The value-transfer allowance is also zero for this declared discrete-time economy. Neither zero is transferred to a diffusion approximation, a constrained nonlinear economy, or an equilibrium calculation.

6.2. Arithmetic and implementation

The verifier encloses (28)–(31) using outward matrix-ball arithmetic. Stored binary coefficients define the model and candidate exactly. Each matrix product encloses rounding and propagated input uncertainty; the code rejects invalid coercivity constants, overflow, and nonfinite entries. It does not use a computed optimal policy. The supplement states the product enclosure and the independent tests.

A separate implementation account allows a state-dependent action error e_t satisfying $\|e_t(x)\|_2 \leq \epsilon_{\text{exec}} \|x\|_2$ at every date. A backward quadratic upper envelope supplies $\varepsilon_{\text{impl}}$, so the final stopping target is

$$\varepsilon_{\text{pol}} + \varepsilon_{\text{impl}} \leq \varepsilon. \quad (32)$$

This is a guarantee for the declared relative-error class, not an assumption that a physical floating-point executor is overflow-free on every real input. The exact stored policy corresponds to zero execution error. The experiment also reports the outward bound for $\epsilon_{\text{exec}} = 10^{-12}$.

6.3. From a trained neural factor to policy accuracy

The preceding verifier accepts any candidate. The next result concerns the *trained representation* itself. For the square-activation continuation $\hat{U}_t(x) = \|W_t x\|_2^2/d + c_t$, every entry of W_t is trainable. This is not a fixed-feature regression assumption. Define its population own-policy derivative loss by

$$\mathcal{L}_t(W_t; P_t) = \frac{1}{4} \|W_t' W_t - d P_t\|_F^2, \quad G_t = \nabla_{W_t} \mathcal{L}_t = W_t(W_t' W_t - d P_t).$$

Here P_t must be evaluated under the *returned* policy, even if the last training target came from an earlier policy. Let $\hat{P}_t = W_t' W_t/d$ and $\hat{P}_T = Q_T$. The residual of the actual actor relative to its fitted continuation is

$$Z_t = (R_t + \beta B_t' \hat{P}_{t+1} B_t) K_t - \beta B_t' \hat{P}_{t+1} A_t.$$

PROPOSITION 7—Trainable-factor certificate: *Suppose $\sigma_{\min}(W_{t+1}) \geq s_{t+1} > 0$. Put $a_{t+1} = \|G_{t+1}\|_F / (ds_{t+1})$ and $a_T = 0$. Then*

$$\|D_t\|_F \leq \|Z_t\|_F + \beta \|B_t\|_2 \|F_t\|_2 a_{t+1}. \quad (33)$$

Replacing $\|D_t\|_F$ in (30) by these upper bounds gives a valid full-policy certificate. A failed rank bound or a large gradient leaves this particular certificate unresolved.

PROOF: Invertibility gives $\|\hat{P}_{t+1} - P_{t+1}\|_F \leq \|G_{t+1}\|_F / (ds_{t+1})$. Direct subtraction shows $D_t = Z_t + \beta B_t' (\hat{P}_{t+1} - P_{t+1}) F_t$. The matrix-norm inequality proves (33); the preceding corollary then applies. *Q.E.D.*

This proposition supplies an a posteriori implication for a jointly trained hidden factor, rather than asserting that a nonconvex optimizer reaches small loss or that every neural architecture has this property. The factor and a conventional quadratic span the same quadratic value class in this experiment. Whether training the factor is worth its additional work is therefore an empirical comparison, not a conclusion of the proposition. The reported factor diagnostic re-evaluates the returned policy, encloses its gradient and rank bound, and retains any unresolved diagnostic. It is a deterministic analysis of the frozen candidates, not a replacement stopping rule selected after observing their performance.

7. A FULL-POLICY CERTIFICATE FOR RECURSIVE RISK PREFERENCES

The order-preserving composition argument does not by itself supply a useful uniform Lipschitz constant for an exponential aggregator on unbounded states. We give a separate, executable bridge for that case. Keep the capital dynamics and vector actions in (26), but evaluate a policy recursively by

$$U_t(x) = \ell_t(x, -K_t x) + \beta \mathcal{R}_\theta U_{t+1}(F_t x), \quad \mathcal{R}_\theta v(y) = \theta^{-1} \log \mathbb{E} e^{\theta v(y + \xi)}, \quad (34)$$

where $F_t = A_t - B_t K_t$, $\theta > 0$, and $\xi \sim N(0, \Sigma)$. The limit at $\theta = 0$ is conditional expectation. This is a recursive entropic criterion with discounting outside the certainty equivalent, not an assertion that it equals every exponential-of-total-cost convention. Exponential-quadratic control and its structural solutions are classical (Jacobson, 1973, Cui et al., 2023). We use that structure to verify a returned neural policy without using an optimal value as a training label or certificate.

For a quadratic continuation $x' P x + c$, Gaussian integration gives

$$\begin{aligned} \mathcal{R}_\theta v(y) &= y' \Psi_\theta(P) y + c + h_\theta(P), \\ \Psi_\theta(P) &= P(I - 2\theta \Sigma P)^{-1}, \quad h_\theta(P) = -\frac{\log \det(I - 2\theta \Sigma P)}{2\theta}. \end{aligned} \quad (35)$$

The sufficient domain condition $2\theta \|\Sigma\|_2 \|P\|_2 < 1$ is checked at every date; a failed domain check is not silently truncated. Own-policy evaluation yields $U_t(x) = x' P_t x + c_t$. Define

$$H_t = R_t + \beta B_t' \Psi_\theta(P_{t+1}) B_t, \quad D_t = H_t K_t - \beta B_t' \Psi_\theta(P_{t+1}) A_t.$$

Choose bounds $\zeta_t \geq \|D_t\|_F$, $f_t \geq \|F_t\|_2$, $b_t \geq \|B_t\|_2$, $p_{t+1} \geq \|P_{t+1}\|_2$, $\sigma \geq \|\Sigma\|_2$, and $\tau \geq \text{tr } \Sigma$. With $\delta_T = \kappa_T = 0$, compute backward

$$\begin{aligned} \gamma_t &= 1 - 2\theta \sigma p_{t+1}, & w_t &= \delta_{t+1} / \gamma_t^2, \\ \bar{f}_t &= f_t + \frac{b_t}{r_t} (\zeta_t + \beta b_t f_t w_t), \\ \delta_t &= \frac{\zeta_t^2}{r_t} + \beta \bar{f}_t^2 w_t, & \kappa_t &= \beta \left(\kappa_{t+1} + \frac{\tau \delta_{t+1}}{\gamma_t} \right). \end{aligned} \quad (36)$$

THEOREM 8—Recursive full-policy comparison: Suppose $Q_t \succeq q_t I \succ 0$, $R_t \succeq r_t I \succ 0$, $Q_f \succeq 0$, and $\Sigma \succeq 0$. If every γ_t is positive and $\delta_t \leq q_t$ for $1 \leq t < T$, then

$$0 \leq U_0(x) - J_0^*(x) \leq \delta_0 \|x\|^2 + \kappa_0 \quad (x \in \mathbb{R}^d). \quad (37)$$

Here J^* compares all adapted policies with finite recursive cost, not only linear policies, and the controls range over all of $\mathbb{R}^d$.

The certificate is a Bellman subsolution: subtract $\delta_t \|x\|^2 + \kappa_t$ from the policy's own value. The matrix and constant allowances are different because the Gaussian log determinant contributes a state-independent cost. The derivative of Ψ_θ supplies γ_t^{-2} , while that of h_θ supplies γ_t^{-1} . These checked quantities replace an uninstantiated global utility Lipschitz constant. The proof and the separately computed execution allowance are in Section S.5.

The verifier encloses own-policy matrices with outward matrix balls. Inverses are proposed numerically and accepted only after a Neumann-residual test. It does not require evaluation of an optimal policy, an enumeration of the Gaussian law, or a state catalogue. In the declared discrete model, class approximation and value-transfer allowances are zero because the comparison is direct; this does not set those allowances to zero for a discretized continuous economy or for the separate strategic models.

For the neural construction, a square-activation critic learns $d\Psi_\theta(P_{t+1})$ after the future policy has been finalized. A separately run structural recursion uses the same analytic information without the factor fit. The comparison therefore isolates the additional neural training cost rather than concealing the structural solution. Neither Gaussian integration nor the classical structural optimum is claimed as a new method. The new conclusion concerns the verified propagation of a returned critic's and policy's errors through a nonlinear recursive preference map.

8. TRAINING TO A RECURSIVE POLICY TARGET

The policy certificate and the training procedure answer different questions. A certificate assesses a returned policy. A constructive training statement must determine an admissible budget before observing that policy's evaluation target.

This section joins the two statements for the trainable square critic in the recursive capital economy of Section 7. The original smooth-network and controlled-diffusion formulations remain unchanged.

8.1. Evaluation without interval expansion

Let $C_T = Q_f$ and let C_t be arbitrary symmetric numerical proposals for the coefficients of the returned policy's own value. They need not be computed by an optimal-control recursion. Write $F_t = A_t - B_t K_t$ and suppose that the following residual and matrix norms are enclosed from above:

$$\rho_t \geq \|C_t - Q_t - K_t' R_t K_t - \beta F_t' \Psi_\theta(C_{t+1}) F_t\|_2.$$

Starting at $e_T = 0$, define

$$\begin{aligned} p_{t+1} &= \|C_{t+1}\|_2 + e_{t+1}, & \gamma_t &= 1 - 2\theta\sigma p_{t+1}, \\ e_t &= \rho_t + \beta \|F_t\|_2^2 e_{t+1} / \gamma_t^2, \end{aligned} \tag{38}$$

where $\sigma \geq \|\Sigma\|_2$. Norms in a numerical implementation mean certified upper bounds, and the computed γ_t is a lower bound.

PROPOSITION 9—Residual-centered evaluation: *If every margin in (38) is positive, the returned policy's recursive value is finite and its exact coefficients satisfy $\|P_t - C_t\|_2 \leq e_t$. Moreover, a valid bound on its full-action residual is*

$$\zeta_t = \|R_t K_t - \beta B_t' \Psi_\theta(C_{t+1}) F_t\|_F + \beta \|B_t\|_2 \|F_t\|_F e_{t+1} / \gamma_t^2. \tag{39}$$

These quantities can be inserted directly into Theorem 8 and its implementation account.

The result is independent of the proposal algorithm. An inaccurate proposal increases its checked Bellman residual; it does not become a valid evaluation merely because it is labeled as one. Propagating a scalar spectral radius around each newly evaluated center avoids feeding an expanding componentwise box through successive matrix inverses. This changes neither the economic criterion nor the stopping tolerance. The certificate remains uniform over real states and full vector deviations, rather than over a stored catalogue.

8.2. A budget chosen from economic primitives

Fix a reference feedback L_t and a positive slack η . This feedback is a feasibility witness, not an optimal solution or a training label. Let $Q_t \succeq q_t I \succ 0$, $R_t \succeq r_t I \succ 0$, and set $q_* = \min_{t < T} q_t$. Construct a scalar coefficient envelope backward:

$$\begin{aligned} \bar{p}_T &\geq \|Q_f\|_2, & \bar{\gamma}_t &= 1 - 2\theta\sigma\bar{p}_{t+1} > 0, \\ \bar{p}_t &\geq \|Q_t + L_t' R_t L_t + \eta Q_t\|_2 + \beta \|A_t - B_t L_t\|_2^2 \bar{p}_{t+1} / \bar{\gamma}_t. \end{aligned} \quad (40)$$

The entire envelope is computed before any policy target or critic fit. For $t < T - 1$, put

$$m_t = dq_{t+1}, \quad M_t^{\text{ub}} = d\bar{p}_{t+1} / \bar{\gamma}_t, \quad f_t = \|A_t\|_2 \left(1 + \frac{\beta \|B_t\|_2^2 M_t^{\text{ub}}}{dr_t} \right).$$

Use the same expression for f_{T-1} with the terminal coefficient bound. Let $b_t \geq \|B_t\|_2$ and $\tau \geq \text{tr } \Sigma$. Define

$$\begin{aligned} \tilde{f}_t &= f_t + \frac{b_t}{r_t} \left(\sqrt{\eta r_t q_t} + \frac{\beta b_t f_t q_*}{\bar{\gamma}_t^2} \right), \\ \lambda_T &= \mu_T = 0, \quad \lambda_t = q_t \mathbf{1}_{\{t < T-1\}} + \frac{\beta \tilde{f}_t^2}{\bar{\gamma}_t^2} \lambda_{t+1}, \\ \mu_t &= \beta \left(\mu_{t+1} + \frac{\tau \lambda_{t+1}}{\bar{\gamma}_t} \right). \end{aligned} \quad (41)$$

For initial states satisfying $\|x_0\|^2 \leq b_0$, write $W_0 = b_0 \lambda_0 + \mu_0$. Choose $s > 0$ such that

$$s \leq \eta, \quad s \lambda_t \leq q_* \quad (1 \leq t < T), \quad s W_0 \leq \varepsilon_{\text{train}}. \quad (42)$$

Zero weights impose no restriction. The remaining allowance $\varepsilon - \varepsilon_{\text{train}}$ is reserved for numerical implementation; it is not presumed sufficient until independently checked.

At date $t < T - 1$, the finalized future policy supplies the own-policy target $M_t = d\Psi_\theta(P_{t+1})$. Train the scalar continuation $\hat{U}_{t+1}(x) = \|W_t x\|^2 / d + \hat{c}_{t+1}$ by

$$W^{(j+1)} = W^{(j)} - \alpha_t W^{(j)} ((W^{(j)})' W^{(j)} - M_t), \quad W^{(0)} = \sqrt{c_t} O_t, \quad (43)$$

where O_t is orthogonal, $0 < c_t \leq m_t$ and $0 < \alpha_t \leq 1/(4M_t^{\text{ub}})$. All entries of W are updated. For $\beta b_t f_t > 0$, the required Gram error is

$$a_t = \frac{d\sqrt{sr_t q_t}}{\beta b_t f_t}, \quad \|(W^{(j)})'W^{(j)} - M_t\|_F \leq a_t. \quad (44)$$

A continuation-independent action needs no such fit. The terminal actor uses the known transformed coefficient $\Psi_\theta(Q_f)$ without a factor fit, as in (S.18). Every actor minimizes its fitted full quadratic action objective, and its own future is then evaluated for the preceding date.

THEOREM 10—Primitive-budget neural policy construction: *Suppose the primitive envelope (40) exists with positive margins. In exact arithmetic, the backward construction just specified terminates after finitely many hidden-weight updates and returns a policy with*

$$0 \leq J_0^\pi(x) - J_0^*(x) \leq s\{\lambda_0\|x\|^2 + \mu_0\} \leq \varepsilon_{\text{train}}, \quad \|x\|^2 \leq b_0. \quad (45)$$

It suffices to take, at each fitted date,

$$k_t = \left\lceil \frac{\log^+(\sqrt{d} M_t^{\text{ub}} / a_t)}{-\log(1 - \alpha_t c_t)} \right\rceil \quad (46)$$

updates. Neither an optimal value nor an already-fitted target is an input to the envelope, tolerance, step size, initialization scale, or cap. The comparator in (45) is the full adapted finite-recursive-cost class in the declared economy.

Section S.6 proves the theorem by joint backward induction on the policy envelope and the critic target. This removes a circular use of a successful fitted policy to justify its own training budget. The conclusion is architecture-specific: it concerns a trainable square factor with a controlled initialization, not a frozen feature dictionary, arbitrary nonconvex initialization, or every network used in the original applications. The class admits an inexpensive structural solution; that comparator remains part of the numerical experiment. Finite neural trainability is not itself a claim that neural training is the least expensive way to solve this class.

The implemented procedure uses conservative caps, retains every failed threshold, and checks the rounded returned policy by Proposition 9. Its final acceptance

rule is the original full-policy tolerance, including execution error. Thus the constructive real-arithmetic theorem does not silently promote a rounded training trace to an economic certificate.

9. WARM-START LEARNING TO A FULL-POLICY ACCURACY TARGET

A local training guarantee is useful for policy construction only when its error allowance is paid in the policy objective. There is a further difficulty with warm starts: their fitted Gram matrices need not underestimate the new continuation. The primitive envelope used for the cold construction cannot therefore be carried over by an appeal to monotonicity. We give an explicit budget that accommodates overshooting factors, errors in own-policy evaluation, and inexact actor solves. The economic model and the comparison with all adapted policies are those of Section 7.

9.1. An implementable local acceptance condition

At date t , write $S_t = \Psi_\theta(P_{t+1}^\pi)$, where the future policy has already been finalized. Suppose an evaluator supplies a symmetric center $\tilde{M}_t$ and a certified radius

$$\|\tilde{M}_t - dS_t\|_2 \leq e_t^M.$$

The fitted scalar continuation has coefficient $\hat{S}_t = W_t'W_t/d$; W_t may be rectangular. Put

$$\begin{aligned} F_t &= A_t - B_t K_t, & Z_t &= R_t K_t - \beta B_t' \hat{S}_t F_t, \\ a_t &\geq \|W_t'W_t - \tilde{M}_t\|_F, & z_t &\geq \|Z_t\|_F. \end{aligned} \tag{47}$$

Thus z_t measures the actor's equation error, not its fitting objective. The exact full-action residual is $D_t = R_t K_t - \beta B_t' S_t F_t$. For any certified bounds $b_t \geq \|B_t\|_2$ and $f_t \geq \|F_t\|_2$,

$$\|D_t\|_F \leq z_t + \frac{\beta b_t f_t}{d} (a_t + \sqrt{d} e_t^M). \tag{48}$$

The two matrix errors enter differently. The Gram error is bounded in Frobenius norm, whereas the evaluation radius is spectral. Omitting the factor $\sqrt{d}$ from the

latter term would not be valid. When an enclosure of $\|F_t\|_F$ is available, it replaces $\sqrt{d}f_t$ in that term.

9.2. A primitive budget that allows overshooting

Keep the reference feedback, slack $\eta > 0$, coefficient envelope $\bar{p}_t$, and positive margins $\bar{\gamma}_t$ from (40). Let

$$q_* = \min_{t < T} q_t, \quad u_t = \bar{p}_{t+1} / \bar{\gamma}_t, \quad v_t = \sqrt{\eta r_t q_*}.$$

All quantities in this paragraph are computed before fitting any target. Define the enlarged closed-loop bound

$$f_t^+ = \|A_t\|_2 \left\{ 1 + \frac{\beta b_t^2 (u_t + q_*)}{r_t} \right\} + \frac{b_t v_t}{r_t}, \quad g_t = f_t^+ + \frac{b_t}{r_t} \left(v_t + \frac{\beta b_t f_t^+ q_*}{\bar{\gamma}_t^2} \right). \quad (49)$$

Norms in a numerical budget denote certified upper bounds. Starting from $\lambda_T^+ = \mu_T^+ = 0$, compute

$$\lambda_t^+ = q_t + \frac{\beta g_t^2}{\bar{\gamma}_t^2} \lambda_{t+1}^+, \quad \mu_t^+ = \beta \left(\mu_{t+1}^+ + \frac{\tau \lambda_{t+1}^+}{\bar{\gamma}_t} \right). \quad (50)$$

Unlike the exact-terminal construction, this account permits an actor error at the last date. For an initial radius $\|x_0\|^2 \leq b_0$, choose $s > 0$ so that

$$s \leq \eta, \quad s \lambda_t^+ \leq q_* \quad (1 \leq t < T), \quad s(b_0 \lambda_0^+ + \mu_0^+) \leq \varepsilon_{\text{policy}}. \quad (51)$$

The local acceptance rule is

$$a_t + e_t^M \leq d q_*, \quad z_t + \frac{\beta b_t f_t^+}{d} (a_t + \sqrt{d} e_t^M) \leq \sqrt{s r_t q_t}. \quad (52)$$

Both inequalities are checked. The first controls the candidate's magnitude; the second pays for its decision error. A positive training tolerance remains only after the evaluator and actor have taken their shares of the budget.

THEOREM 11—Budgeted warm-start policy construction: *Suppose the primitive coefficient envelope has positive moment margins and (51) holds. Construct the policy*

backward. At every date, evaluate the finalized future policy, return a positive-semidefinite fitted coefficient and an actor, and verify (47)–(52). Then $0 \preceq P_t^\pi \preceq \bar{p}_t I$ and

$$0 \leq J_0^\pi(x) - J_0^*(x) \leq s\{\lambda_0^+ \|x\|^2 + \mu_0^+\} \leq \varepsilon_{\text{policy}}, \quad \|x\|^2 \leq b_0. \quad (53)$$

The comparison is with all adapted policies of finite recursive cost, over all real vector controls. No commutation or one-sided approximation condition on the fitted Gram matrices is required.

For a warm fit whose center, initial factor, step error, and precision satisfy Theorem S.8, a finite cap follows by substituting its positive remaining Gram allowance in (S.22). The evaluator’s radius and the actor equation error must still satisfy (52). A training gate alone is not the acceptance rule for the economic policy.

The proof in Section S.10 jointly establishes the coefficient envelope and the full-policy bound. The extra q_* in (49) pays for approximation on either side of the target; the final term pays for the actor solve. The construction does not use an optimal policy as a training label. At the last date the true continuation coefficient is $\Psi_\theta(Q_f)$, not Q_f unless $\theta = 0$ (or the transform is otherwise unchanged). The terminal target may be supplied directly and its actor checked without training another continuation.

COROLLARY 12—A finite arithmetic implementation: *Suppose the economic matrices, risk parameter, discount factor, and supplied bounds are rational, and the stated moment margins are strict. Own-policy coefficient evaluation and actor equations can be solved by exact rational matrix arithmetic. At a certified warm start, compute the factor update exactly and round its entries to a dyadic grid. Choosing the grid fine enough that its Gram noise floor is below the remaining positive allowance gives a finite number of updates satisfying (52). When a warm gate fails, the controlled cold construction with rational squared initialization provides a finite alternative at nonterminal dates. Consequently the policy bound has a finite-arithmetic realization, rather than only a formal real-arithmetic interpretation.*

This corollary is a statement about a specified arithmetic routine. Ordinary floating-point matrix products require their own full-step error enclosure; nearest-

storage rounding by itself does not cover them. The independent rounded-policy verifier is retained for the larger numerical experiments. An additional action-execution allowance is added to (53) before comparing with a requested tolerance.

9.3. Construction, reliability, and work

For the declared admissible inputs, the deterministic cap is uniform over warm factors and permitted update errors. It is not a reliability interval estimated from a few fitting seeds. Its economic content is a finite construction to a full-policy tolerance under recursive preferences, including the cost of precision. The exact-arithmetic example in the replication package checks the policy envelope, non-commuting Gram errors, evaluation, and the returned vector actors against the same rational inputs. It is a constructive validation instance, not an additional independent sample in the earlier economic experiments.

An amortization comparison includes evaluating the changed future, testing the warm gate, unsuccessful checks, fallback training, actor solves, verification, and durable output. The theory gives a cap for the neural factor updates; it does not turn that cap into a lower wall-clock cost than an analytic structural solution. The earlier cost frontiers and their unfavorable comparisons are retained. In particular, improving the trainable continuation’s mathematical guarantee does not change which procedure was cheapest in an already executed experiment.

10. PRECISION-ADAPTIVE NEURAL CONTINUATION REFRESH

An old continuation can be inaccurate for a changed economy and nevertheless provide a useful initialization for learning its new continuation. A useful refresh rule must specify both how much learning is required and how accurately that learning must be implemented. We give such a rule for the trainable square continuation, allowing rectangular factors and noncommuting targets. The economic stopping criterion remains the full-policy bound of Theorem 11.

Fix one date after the future policy has been finalized. During this fit the symmetric target M is fixed and satisfies $mI \preceq M \preceq LI$, with $0 < m \leq L$. For $W_j \in \mathbb{R}^{p \times d}$,

$p \geq d$, let

$$C_j = M^{-1/2} W_j' W_j M^{-1/2}, \quad E_j = C_j - I.$$

The implemented update is

$$W_{j+1} = \frac{1}{2} W_j \{3I - M^{-1} W_j' W_j\} + \Delta_j, \quad \|\Delta_j M^{-1/2}\|_2 \leq v_j. \quad (54)$$

Every entry of the hidden factor is trainable. The inverse is that of the current fixed target; it is neither an inverse of the optimal value nor a matrix transported unchanged from a different economic future. In whitened coordinates, the ideal update is a Newton–Schulz polar iteration. Matrix polar iterations and their numerical analysis are classical (Higham, 2008). We use their error identity to allocate precision within an own-policy economic error budget, rather than claim a new matrix iteration.

Choose a checked basin radius $0 < r \leq 1/2$ and put $c_r = (3 + r)/4$. If $\|E_0\|_2 \leq \rho_0 \leq r$, define $\rho_{j+1} = \rho_j^2$. For every step that is needed, require

$$2v_j + v_j^2 \leq (1 - c_r) \rho_j^2. \quad (55)$$

A zero initial error needs no update. The condition budgets the whole implemented step. It cannot be justified by bounding only the final storage conversion when preceding matrix products have also been rounded.

THEOREM 13—Precision-adaptive refresh: *Suppose the fixed target, initial basin, and full-step errors satisfy the conditions just stated. Then every returned factor obeys*

$$(1 - \rho_j)M \preceq W_j' W_j \preceq (1 + \rho_j)M, \quad \rho_j = \rho_0^{2^j}. \quad (56)$$

For a requested relative tolerance $0 < \tau < 1$ with $\tau < \rho_0$, a sufficient number of updates is

$$K = \left\lceil \log_2 \left\{ \frac{\log(1/\tau)}{\log(1/\rho_0)} \right\} \right\rceil. \quad (57)$$

The number is zero if $\rho_0 \leq \tau$. No simultaneous eigenbasis for $W_0' W_0$ and M is assumed.

Suppose, more specifically, that matrix products and target solves are exact and each of the pd entries of the updated factor is rounded to the nearest multiple of 2^{-b_j} . Let A be any

finite certified upper bound for $\sqrt{pd/m}$. Choosing the first positive integer b_j for which

$$2v_j + v_j^2 \leq (1 - c_r)\rho_j^2, \quad v_j = A2^{-b_j-1}, \quad (58)$$

is sufficient. Such an integer exists at every required step. For a constant C depending only on A and r , the chosen integers satisfy

$$\sum_{j=0}^{K-1} b_j \leq CK + 4\log_2(1/\tau). \quad (59)$$

A checked inverse of the fixed target may be constructed once and applied at each update. Thus a fit requiring updates needs one target-inverse construction and K target-inverse applications; a zero-update fit needs neither.

The proof is in Section S.11. The first conclusion is a uniform deterministic statement over the declared basin and perturbations. The second is a fractional-storage account, not a bound on all bit operations in exact rational arithmetic. It omits neither inverse applications nor residual and positivity checks from a measured service cost. Numerators and denominators in exact intermediate products can be considerably larger than the stored factor precision. An implementation with a fixed precision cap must fail explicitly if it cannot satisfy (58); the theorem does not authorize it to return an uncertified factor.

10.1. A target change and its economic allowance

Suppose an old factor has relative error at most ρ_{old} for M_{old} , and evaluation of the new future establishes

$$(1 - \omega)M_{\text{new}} \preceq M_{\text{old}} \preceq (1 + \omega)M_{\text{new}}, \quad 0 \leq \omega < 1.$$

Then the same factor has relative error at most $\rho_{\text{old}} + \omega + \rho_{\text{old}}\omega$ for M_{new} . If this upper bound does not exceed r , it supplies an admissible initial radius for Theorem 13. Otherwise, a direct basin test or a controlled cold construction is required. This is a condition for using an old factor as an initialization, not for accepting its old policy without improvement. The old inverse is not reused across the target change.

COROLLARY 14—Precision-budgeted full-policy construction: *Under the primitive envelope and allocation of Theorem 11, let each positive remaining Gram allowance be a_t , and let the independently evaluated center satisfy $m_t I \preceq \tilde{M}_t \preceq L_t I$. A basin-certified refresh with*

$$\tau_t \leq \min\{1/2, a_t / (\sqrt{d} L_t)\} \quad (60)$$

reaches that Gram allowance after the finite schedule in (57)–(58). If the evaluation radius and returned actor also satisfy (52), then

$$0 \leq J_0^\pi(x) - J_0^*(x) \leq s\{\lambda_0^+ \|x\|^2 + \mu_0^+\} \leq \varepsilon_{\text{policy}}, \quad \|x\|^2 \leq b_0.$$

The comparison is with all adapted policies of finite recursive cost and all real vector controls in the stated economy. A separate action-execution allowance is added when the implemented actions differ from the stored rational feedback.

Thus the economic primitives determine the envelope and policy allocation; own-policy evaluation supplies the current target and its radius; the precision rule constructs a trainable continuation to the remaining allowance; and the actor and full-policy checks complete the certificate. The terminal target is the known transformed coefficient $\Psi_\theta(Q_f)$, not Q_f in general. A direct terminal evaluation requires no neural factor fit.

The update count and fractional precision account concern a specified neural construction. They do not rank all ways of solving the economy. In particular, the structural solution legitimately bypasses factor training. The numerical comparison below charges both constructions to the same economic stopping target, includes that competitor, and reports the work of evaluating, checking, and writing the result as well as the work of training.

11. CONSTRUCTIVE NEURAL BELLMAN OPERATORS BEYOND QUADRATIC CONTINUATIONS

A continuation method has an economic use when its approximation can be translated into a decision guarantee without assuming that the approximation class contains the value function. We now give this translation for nonlinear recursive economies. The comparison is with all feasible adapted policies. The construction

uses compact state and action sets, not a Gaussian coefficient identity. The preceding diffusion framework and unbounded-state results remain in force under their respective assumptions; compactness is imposed only in this additional realization.

11.1. A residual-oscillation theorem

Let X and A be compact metric spaces, let $F_t : X \times A \times Z \rightarrow X$ and $c_t : X \times A \rightarrow \mathbb{R}$ be continuous, and let Z have a finite distribution with fixed probabilities q_z . Write $\beta_t > 0$ for the date- t continuation weight, $B_0 = 1$, and $B_t = \prod_{j < t} \beta_j$. The conditional continuation functional ρ_t is monotone and cash invariant: $\rho_t(Y + a) = \rho_t(Y) + a$. It is continuous on finite vectors. In particular, these conditions hold for

$$\rho_\theta(Y) = \theta^{-1} \log \sum_z q_z e^{\theta Y(z)}, \quad \rho_0(Y) = \sum_z q_z Y(z), \quad \theta \geq 0. \quad (61)$$

Boundedness makes the exponential moment finite. Define

$$Q_t(v; x, a) = c_t(x, a) + \beta_t \rho_t(v(F_t(x, a, Z))), \quad T_t v(x) = \min_{a \in A} Q_t(v; x, a).$$

Continuity gives measurable minimizers. Recursive costs for arbitrary adapted policies use the same conditional functional at every realized history. Let $V_T^* = g$ and $V_t^* = T_t V_{t+1}^*$.

THEOREM 15—Nonlinear full-policy comparison: *Let v_t be continuous functions and let $\pi_t(x) \in A$ be measurable. Suppose that, at every state,*

$$\ell_t \leq v_t - T_t v_{t+1} \leq u_t, \quad t < T, \quad (62)$$

$$Q_t(v_{t+1}; x, \pi_t(x)) - T_t v_{t+1}(x) \leq \eta_t, \quad (63)$$

$$\ell_T \leq v_T - g \leq u_T.$$

Then, for every initial state and against all feasible adapted policies,

$$0 \leq J_0^\pi - V_0^* \leq \sum_{t < T} B_t(u_t - \ell_t + \eta_t) + B_T(u_T - \ell_T). \quad (64)$$

Moreover, $v_0 - \sum_{t \leq T} B_t u_t$ is a lower bound on V_0^ . Neither membership of V^* in a neural class nor convergence of a training optimizer is required for this comparison.*

PROOF: Monotonicity and cash invariance imply $\|T_t v - T_t w\|_\infty \leq \beta_t \|v - w\|_\infty$; the same statement holds for the operator of any fixed feasible action. Backward induction applied to (62) yields

$$v_t - \sum_{s=t}^T (B_s / B_t) u_s \leq V_t^* \leq v_t - \sum_{s=t}^T (B_s / B_t) \ell_s.$$

The terminal equality is immediate. For the induction step, shift the next function by the corresponding deterministic constant, apply T_t , and use $T_t v_{t+1} \in [v_t - u_t, v_t - \ell_t]$. For the returned policy, (63) instead gives

$$J_t^\pi \leq v_t + \sum_{s=t}^{T-1} (B_s / B_t) (\eta_s - \ell_s) - (B_T / B_t) \ell_T.$$

Subtract the lower bound on V_t^* . At an arbitrary history, each realized action has continuation cost at least the Bellman minimum, and induction compares its future cost with V_{t+1}^* . Thus the Bellman value is a lower bound for every adapted policy, not only for the Markov or neural class. A measurable minimizing Markov policy attains it. This proves both claims. *Q.E.D.*

The economically relevant residual is its oscillation $u_t - \ell_t$. A common additive valuation error cancels from the loss comparison. This is the nonlinear counterpart of centered continuation error in the earlier operator analysis, rather than an assumption that a critic learns absolute value levels accurately.

11.2. A finite construction and a trained-network check

Take $X = [0, 1]$ and $A = [0, \bar{a}]$. Suppose c_t and F_t have known Lipschitz constants separately in state and action. If v_{t+1} has Lipschitz constant L_{t+1} , admissible constants for Q_t are

$$L_{x,t} = L_{c,x,t} + \beta_t L_{F,x,t} L_{t+1}, \quad L_{a,t} = L_{c,a,t} + \beta_t L_{F,a,t} L_{t+1}. \quad (65)$$

The same $L_{x,t}$ bounds the state variation of $T_t v_{t+1}$. Use a uniform state partition $x_i = ih$ and an action grid of mesh Δ , including both endpoints. At every pair, verified arithmetic supplies $\underline{q}_{ij} \leq Q_t(v_{t+1}; x_i, a_j) \leq \bar{q}_{ij}$. Set $L_i = \min_j \underline{q}_{ij}$, $U_i = \min_j \bar{q}_{ij}$

and $\delta_t = L_{a,t}\Delta/2$. Let v_t be the linear interpolant of stored nodal values y_i , with no requirement that these equal an exact minimizer. Then valid residual constants are

$$\ell_t = \min_i (y_i - U_i) - L_{x,t}h/2, \quad (66)$$

$$u_t = \max_i (y_i - L_i + \delta_t) + L_{x,t}h/2.$$

If the action a_{j_i} is stored at node i and extended by the nearest-node rule, a valid actor allowance is

$$\eta_t = \max_i (\bar{q}_{ij_i} - L_i) + \delta_t + L_{x,t}h. \quad (67)$$

All expressions are enclosed outward in the implementation.

PROPOSITION 16—Constructive nonquadratic continuation: *Under the preceding conditions, (66)–(67) and a certified terminal interpolation error imply Theorem 15. Every such v_t has the exact representation*

$$v_t(x) = y_0 + s_0x + \sum_{i=1}^{N-1} (s_i - s_{i-1})(x - x_i)_+, \quad s_i = (y_{i+1} - y_i)/h. \quad (68)$$

For each fixed finite horizon, a feasible neural policy with any prescribed positive loss tolerance can be constructed in finitely many operations if arithmetic precision can be increased to deliver the required interval widths. This assertion concerns the constructive mesh algorithm, not unrestricted stochastic-gradient training.

PROOF: The continuous action minimum at a node lies in $[L_i - \delta_t, U_i]$: a grid point is within $\Delta/2$ of any minimizing action. On a state cell, linear interpolation of nodal Bellman values differs from the Bellman function by at most $L_{x,t}h/2$, since the interpolation-weighted distance to the two endpoints is $2r(h-r)/h \leq h/2$. This proves (66). Moving from a state to its nearest node changes both its chosen-action value and the Bellman minimum by at most $L_{x,t}h/2$ each, proving (67). Slope changes prove (68).

For finiteness, choose nodal midpoints of $[L_i, U_i]$ and the action minimizing $\bar{q}_{ij}$. Suppose each Q interval has width at most ω and each stored midpoint has an additional rounding error at most ζ . Then $U_i - L_i \leq \omega$. Residual oscillation plus

actor allowance is at most

$$2\omega + 2\zeta + 2\delta_t + 2L_{x,t}h.$$

For the Lipschitz recursion, each nodal midpoint is within $\delta_t + \omega + \zeta$ of the exact Bellman value. Consequently the next interpolant has Lipschitz constant at most $L_{x,t} + 2(\delta_t + \omega + \zeta)/h$. Choose $\Delta \leq h^2$ and $\omega + \zeta \leq h^2$ at every date. Equations (65) then give, by finite backward induction, a bound on all interpolant Lipschitz constants uniform over $0 < h \leq 1$. Terminal interpolation contributes at most $L_g h/2 + \zeta$ in absolute error. The right-hand side of (64) therefore tends to zero with h . A sufficiently fine finite grid and sufficiently precise finite arithmetic attain any positive target. This proof does not assert that fixed binary64 arithmetic suffices for every target. Q.E.D.

A conventional spline routine and (68) represent the same constructed function; relabeling it alone is not a neural work advantage. The second implementation trains all affine hidden weights in $f_t(x) = a_t + b_t x + \sum_{j=1}^p c_{tj}(w_{tj}x + d_{tj})_+$. It uses the certified nodal values as deterministic training data and selects actions from its own continuation predictions. Its selected actions are then checked by (67) against the independent interval brackets. The full cost of the reference construction, training and verification is charged. Acceptance certifies the returned policy even when the optimizer has no convergence guarantee. A rejected proposal can be replaced by the constructive policy above; this gives a terminating certified procedure without asserting that every trained proposal must be accepted.

PROPOSITION 17—All-state verification of trained ReLU continuations: *Let f be the preceding one-hidden-layer scalar ReLU network and v a linear spline. The maximum of $|f - v|$ on $[0, 1]$ occurs at an endpoint, a spline knot, or a neural root $-d_j/w_j$ in that interval, with $w_j \neq 0$. Enclosing each root by outward division and evaluating both functions over its interval gives a certified uniform bound, including nondyadic roots. Direct evaluation costs $O((N + p)(p + \log N))$ arithmetic operations; sorting the breakpoints, when used, costs $O((N + p) \log(N + p))$.*

PROOF: Between consecutive points in the specified set, $f - v$ is affine. The absolute value of an affine function is convex and is bounded by its endpoint maximum. Continuity includes the endpoints of adjacent pieces. Zero hidden slopes produce no breakpoint. Every exact breakpoint belongs to its outward root interval, so interval evaluation bounds its value even when a root cannot be stored exactly. A direct network evaluation uses $O(p)$ operations and a binary-search spline evaluation $O(\log N)$. There are at most $N + p + 1$ points. *Q.E.D.*

This certificate removes the mesh-Lipschitz remainder used in the earlier network-to-spline check. It does not retroactively change the earlier policy bounds or execution clocks. The new endpoint audit uses exact rational arithmetic only as an independent check of the verifier, not as the production method used in the floating-point economic services.

11.3. Nonlinear investment and the value of reoptimization

The economic realization has bounded capital, irreversible investment and a shortfall penalty. Its primitives are

$$F(x, a, z) = \text{clip}_{[0,1]} \{13x/16 + a + x(1 - x)/16 + z\}, \quad (69)$$

$$c_P(x, a) = (x - 11/16)^2 + Pa^2 + 4a^4 + 2(3/8 - x)_+^2,$$

$$g(x) = 2(x - 11/16)^2 + 2(3/8 - x)_+^2, \quad \beta = 15/16,$$

where $a \in [0, 1/4]$ and the shocks $(-1/16, 0, 1/16)$ have probabilities $(1/4, 1/2, 1/4)$.

The price P measures the quadratic investment burden; θ in (61) measures aversion to high future costs. A low-capital household faces both a distance-to-target loss and a shortfall penalty. Investment is capped and cannot be negative. The quartic adjustment cost, nonlinear accumulation and clipping prevent a Riccati reduction. This is a theoretical counterfactual economy, not an estimated calibration.

Valid primitive constants are $L_{c,x} = 23/8$, $L_{F,x} = 7/8$, $L_{c,a} = P/2 + 1/4$, $L_{F,a} = 1$, and $L_g = 17/4$. They follow by differentiation on smooth pieces and continuity at the kinks; clipping is nonexpansive. Hence Proposition 16 verifies its assumptions directly from the primitives.

PROPOSITION 18—Fixed-rule investment exposure: *For a fixed feasible policy rule π in (69), independent of P , the derivative $D_t^\pi = \partial J_t^\pi / \partial P$ exists along each finite shock tree and satisfies $D_T^\pi = 0$ and*

$$D_t^\pi = a_t^2 + \beta \sum_z \tilde{q}_{t,z} D_{t+1}^\pi(F_t), \quad \tilde{q}_{t,z} = \frac{q_z \exp(\theta J_{t+1}^\pi(F_t))}{\sum_y q_y \exp(\theta J_{t+1}^\pi(F_t))}. \quad (70)$$

At $\theta = 0$, $\tilde{q} = q$. In particular $D_t^\pi \geq 0$. For old and new rules, the total cost change decomposes exactly as

$$J_0^{\pi^{\text{new}}}(4) - J_0^{\pi^{\text{old}}}(1) = \{J_0^{\pi^{\text{old}}}(4) - J_0^{\pi^{\text{old}}}(1)\} - \{J_0^{\pi^{\text{old}}}(4) - J_0^{\pi^{\text{new}}}(4)\}. \quad (71)$$

The first bracket is mechanical repricing; the second is the cost saved by reoptimization at the new price.

PROOF: For a fixed rule, the states and selected actions on the finite shock tree do not depend on P , even if the rule is discontinuous in state. Each stage cost is affine in P . Differentiate the finite log-sum-exp recursion; its derivative is the probability-weighted derivative in (70). Backward induction proves existence and nonnegativity. Add and subtract $J_0^{\pi^{\text{old}}}(4)$ to obtain (71). *Q.E.D.*

No global monotonicity of the optimal investment rule is inferred from this fixed-rule statement. The numerical counterfactual below instead reports the actual policy changes and an independent lower bound on the second bracket. A strictly positive lower bound establishes an economic gain from the returned policy even when its optimality certificate is conservative.

12. VERIFIED FLOATING-POINT IMPLEMENTATION AND COMPLETE WORK

The factor iteration is useful only if its whole-step allowance can be obtained for the operations actually performed. The next result supplies such an allowance without exact matrix products, exact solves, or a claim that a small computed residual is automatically rigorous. The residuals themselves must be enclosed.

PROPOSITION 19—Residual verification of a native update: *Let M be the fixed stored symmetric target with a proved bound $M \succeq mI$, $m > 0$, and let $W \in \mathbb{R}^{p \times d}$. Let Z and V be arbitrary stored floating-point proposals. Define their exact-real residuals by*

$$R = MZ - W^\top W, \quad D = V - \frac{1}{2}W(3I - Z).$$

If outward arithmetic proves $\|R\|_F \leq r_R$, $\|D\|_F \leq r_D$, and $\|W\|_2 \leq w$, then the ideal update $\bar{W} = \frac{1}{2}W(3I - M^{-1}W^\top W)$ satisfies

$$\|(V - \bar{W})M^{-1/2}\|_2 \leq \nu := \{r_D + wr_R/(2m)\}/\sqrt{m}. \quad (72)$$

Writing $E = M^{-1/2}W^\top WM^{-1/2} - I$, if $\|E\|_2 \leq r < 1$, then

$$\|M^{-1/2}V^\top VM^{-1/2} - I\|_2 \leq (3 + r)r^2/4 + 2\nu + \nu^2. \quad (73)$$

Thus $2\nu + \nu^2 \leq (1 - r)r^2/4$ certifies a squared-radius update. Alternatively, a direct Gram enclosure can certify a terminal target without certifying this contraction.

PROOF: $Z - M^{-1}W^\top W = M^{-1}R$, so $V - \bar{W} = D - \frac{1}{2}WM^{-1}R$. Submultiplicativity gives (72). Put $Y = WM^{-1/2}$ and $\bar{Y} = \bar{W}M^{-1/2}$. Then $\bar{Y} = \frac{1}{2}Y(3I - Y^\top Y)$ and $\bar{Y}^\top \bar{Y} - I = (-3E^2 + E^3)/4$. For each eigenvalue $e \in [-r, r] \subset (-1, 1)$, this polynomial is nonpositive, and $(1 + e)(1 - e/2)^2 \leq 1$. Hence $\|\bar{Y}\|_2 \leq 1$. The implemented perturbation adds at most $2\nu + \nu^2$ to the Gram error, proving (73). The argument works for every $r < 1$; it does not apply the earlier $r \leq 1/2$ budget outside its stated range. Q.E.D.

The implementation forms proposals in binary32 or binary64 and verifies them with outward binary64 matrix balls. A cached approximate inverse is permitted only for the identical fixed target and precision. Residual R is evaluated against the original stored target and factor, so conversion and solve errors, including those of the cached inverse, enter the enclosure. A failed binary32 proposal can be retried in binary64. Exhausting the available native types is reported as a failure, not as a proof that higher precision would fail. Positive spectral bounds are certified by sufficient Gershgorin inequalities; matrices failing this sufficient check are not silently accepted on the basis of a computed eigenvalue. The experiments concern

exact-real feedback rules with stored binary coefficients. Floating-point execution of the actor needs the separate execution allowance already present in the policy theorem.

12.1. *Work and storage accounting*

Let n_b be the number of target-inverse constructions, n_a the number of proposed factor updates including rejected proposals, n_e the number of complete own-policy evaluations, n_c the number of complete policy checks, and n_h the number of actor linear solves. For dense matrices, a sufficient arithmetic-operation account for this implementation is

$$O\left(n_b d^3 + n_a(p d^2 + d^3) + (n_e + n_c) T d^3 + n_h d^3 + \mathcal{W}_{\text{io}}\right). \quad (74)$$

Here $\mathcal{W}_{\text{io}}$ includes recorded serialization and output work. The ball products and residual checks have the same dense-operation order as the proposed update, but their constants are not those of a single BLAS call. A working storage bound, excluding the accumulated audit log, is $O(T(p d + d^2) + p d + d^2)$ words. The stored factor remains binary64 even when its proposal was calculated in binary32. Reduced proposal precision therefore does not imply reduced persistent factor storage in this implementation.

For an accepted fixed-target contraction path, a sufficient update cap is $K = \lceil \log_2(\log(1/\tau)/\log(1/r_0)) \rceil_+$. An ideal harmonic scalar initialization has $r_0 = (\kappa - 1)/(\kappa + 1)$, where $\kappa = L/m$; the implemented initial radius includes rounding and is checked separately. At most two native proposal attempts are made per accepted step when both types are available. Conditioning also enters (72) through m^{-1} and $m^{-1/2}$. The economic target determines τ through the previously derived actor, evaluation and Gram allowances; the entropic moment margin enters those allowances rather than disappearing into an iteration count. If a gate fails, the realized counters, not K alone, determine complete work. Equation (74) is an arithmetic-operation bound at the stated native precisions, not a total arbitrary-precision bit-complexity theorem.

For the scalar nonlinear construction, $N + 1$ states, $A + 1$ actions and s shocks require $O(TNAs \log N)$ operations with binary-search spline evaluation, in addition to the cost of certified transcendental evaluation. A sweep over sorted successor states can change this bound but is not presumed here. All fixed-rule shock-tree evaluations are also charged; their worst-case size is $O(s^T)$. Neither this finite construction nor the endpoint verifier asserts dimension-independent approximation complexity.

12.2. Relation to numerical methods

Classical polar and matrix-function iterations supply the algebraic update, not its economic interpretation. Mixed-precision iterative refinement separates working, correction and residual precision; [Carson and Higham \(2018\)](#) provides a systematic analysis of that separation. Verified numerical computation, as surveyed by [Rump \(2010\)](#), makes the additional distinction between a computed residual and an enclosing bound. Our residual construction uses both ideas, while requiring neither to imply an economic policy guarantee by itself. That guarantee comes from the Bellman comparison and its allocation of evaluation, actor and continuation error. Likewise, the recursive-risk quadratic comparator exploits the structural control solution, whereas Section 11 does not use a quadratic closure. The new contribution is the connection between certified numerical work and a returned policy in the original NBO framework, not a claim to have invented polar iteration, spline approximation, or residual verification.

13. IDENTIFIED COMPARISONS AND A NONLINEAR ECONOMIC COUNTERFACTUAL

The original exact-construction cases remain below, including the faster structural comparator and the earlier adverse reuse comparisons. They are not used to identify a precision effect. We instead analyze the deposited R38 factorial and nonlinear execution, and add an independently checked all-state network certificate. No historical clock is replaced by a reconstruction.

13.1. *Design and accounting*

The fixed protocol contains 315 isolated services: 288 quadratic policy services, 12 spline services, 12 trained-network services and three nonlinear accuracy-frontier services. There are 105 deterministic economy–method objects with three timing repetitions each. A separate warm-up is excluded from the contrasts. Execution order is shuffled with a fixed seed, and the numerical libraries use one thread. Neither CPU affinity nor frequency was controlled. The displayed time is the complete child service through output synchronization; the archive also records startup-inclusive parent time. Minima and maxima accompany medians rather than a significance claim from three observations. Memory is process high-water RSS, not exclusive algorithm memory.

The quadratic design crosses fixed binary64 versus adaptive native precision with inverse rebuilding versus within-target caching. It adds a cached fixed method selected from binary32 and binary64, in that order, and the structural solution. All tuning trials, failed policy checks and precision retries enter the service boundary. The selected fixed type is the lowest successful type in that two-element set, not the minimum possible arbitrary mantissa length. The update rule is held fixed in this factorial; comparisons with the older update rules remain separately identified in the historical table.

Every candidate policy is checked initially and after an actor change; the service stops at its first successful economic certificate. The target grid is $10^{-2}, 10^{-4}, 10^{-6}, 10^{-8}, 10^{-10}$. Further objects vary (d, T) over $(8, 4), (16, 8), (32, 4), (4, 12)$ with condition parameters 8, 16, 16, 4, change the valuation by factors 1.005, 1/4, 4, include cold and transported starts, and include an additional moment-margin case. This finite design tests several operating regimes; it is not an asymptotic scaling experiment or an exhaustive stress test near the entropic moment boundary.

The factorial removes the original inverse-caching confound. It does not show a stable timing advantage from adaptive precision. At tolerance 10^{-4} , the fixed64 cached median is about 0.415 seconds and the adaptive cached median about 0.433 seconds; the structural median is about 0.170 seconds. The two cached factor pro-

TABLE II
PRECISION AND CACHING AT A COMMON POLICY TARGET

Method	Median (s)	Range (s)	Updates	Builds	Loss bound
Fixed64, rebuilt	0.416	[0.416, 0.420]	11	11	5.59×10^{-6}
Fixed64, cached	0.415	[0.415, 0.421]	11	3	5.59×10^{-6}
Adaptive, rebuilt	0.421	[0.418, 0.422]	11	11	5.59×10^{-6}
Adaptive, cached	0.433	[0.417, 0.433]	11	3	5.59×10^{-6}
Tuned native, cached	0.424	[0.423, 0.428]	11	3	5.59×10^{-6}
Structural	0.170	[0.168, 0.173]	0	0	4.46×10^{-28}

$d = 2$, $T = 4$, target 10^{-4} . Three isolated complete-service clocks per object. Builds count target inverses, not actor solves. All recorded failed checks and tuning are charged.

cedures each make eleven updates and three inverse constructions. The uncached procedures construct eleven inverses. Reusing an inverse changes this count, but verification and own-policy evaluation can dominate the small-matrix service clock.

TABLE III
FIRST CERTIFIED POLICY ACROSS ACCURACY TARGETS

Target	Adaptive bound	Bound/target	Adaptive (s)	Fixed64 (s)	Structural (s)
0.01	0.0023	0.23	0.352	0.353	0.175
0.0001	5.59×10^{-6}	0.0559	0.433	0.415	0.170
1×10^{-6}	4.68×10^{-11}	4.68×10^{-5}	0.444	0.442	0.171
1×10^{-8}	4.35×10^{-11}	0.00435	0.500	0.482	0.169
1×10^{-10}	4.35×10^{-11}	0.435	0.486	0.491	0.175

Factor methods use caching. Times are medians; complete dispersion and all methods appear in the supplement. Discrete oversolving is displayed, not suppressed.

First-pass stopping removes unnecessary continuation after a successful policy check. It does not force a discrete update to land at the target. The achieved bound-to-target ratio for adaptive caching ranges from approximately 4.68×10^{-5} to 0.435 over these five targets. In particular, the 10^{-6} object still oversolves substantially. Table III therefore reports achieved error as well as work; it is not presented as a frontier with uniformly binding constraints. The structural solver also oversolves,

for a different reason: the economy permits an exact structural policy. All 288 policy services certify their declared targets. Complete counters, process memory and timing dispersion for every cell are supplied in the supplement and the machine-readable audit.

Small and large transported changes have different operational consequences. For adaptive caching, the 1.005 change uses three new factor updates rather than twelve from a cold start. The factor-4 change fails all three transported factor gates and falls back to cold construction. The factor-1/4 change also fails the transported gates. Setup of the old policy is charged jointly in the reported warm service; accordingly its total clock need not beat a cold solve. These are observed gate outcomes, not a success rate estimated for an unknown population of future changes.

13.2. *Nonlinear approximation, policy accuracy and economic gains*

The economy in Section 11.3 is run for four decision dates at $(P, \theta) \in \{1, 4\} \times \{0, 1\}$. The direct construction uses $N = 256$ state cells and $A = 128$ action cells. The trained continuation has 31 hidden units, changes all affine hidden weights, and uses 300 Adam steps per date. Training data are deterministic certified Bellman nodal values, not independent observations. Every state and every continuous feasible action are covered by the interpolation and action allowances in the theorem. The finite shock sum is evaluated through interval operations with Taylor enclosures for exponential and logarithm; no analytic Gaussian continuation is used.

TABLE IV

NONLINEAR CAPITAL: POLICY BOUNDS AND COMPLETE WORK

Price	Risk	Neural bound	Spline bound	Neural (s)	Spline (s)
1	0	0.2359	0.2352	2.846	0.240
1	1	0.2366	0.2359	5.011	1.311
4	0	0.2533	0.2520	2.999	0.409
4	1	0.2542	0.2528	6.222	2.564

$T = 4$, $N = 256$, $A = 128$. Cost includes reference construction and, for neural services, all fitting and verification. Price-four services also include construction and evaluation of the old rule. Bounds are in model cost units.

The trained policies have global loss bounds between approximately 0.236 and 0.254. These are deliberately reported in the model’s cost units, not interpreted as 10^{-4} certificates. The spline method is faster in all four complete-service comparisons. A separate constructive accuracy sweep has first-pass bounds 0.9352, 0.4704, 0.2359, 0.1181 at targets $1, 1/2, 1/4, 1/8$, respectively, using $N = 64, 128, 256, 512$ and $A = N/2$. The corresponding bound-to-target ratios lie between 0.935 and 0.946. Its cumulative construction clocks include the unsuccessful coarser meshes; they are reported separately from the full service clock and are not four independent experimental observations from each repetition.

The price change nevertheless supports a substantive policy comparison. The old rule is the certified conventional rule constructed at $P = 1$ and then held fixed; the new rule is selected by the trained neural continuation at $P = 4$. Both are evaluated under the new price, by an independent finite-shock tree enclosure. Table V reports the second bracket in (71). Its lower bound is positive at each reported initial state for both risk specifications. Thus the new policy strictly improves on retaining the old rule under the changed investment burden. This comparison is not a claim that the neural rule outperforms the newly optimized spline rule. The latter remains the stronger work comparator.

Higher investment burden is accompanied here by lower initial investment in low and intermediate capital states. The fixed-rule exposure formula explains why keeping the old action rule incurs a positive repricing burden, while the independent policy comparison quantifies how much of that burden is avoided. The gain is economically interpretable even though the uniform global optimality bound is more conservative. It does not identify an empirical elasticity or a calibrated welfare effect.

13.3. *New all-state network verification*

The endpoint verifier in Proposition 17 is applied to all 60 deposited date-specific network records. Deduplication by coefficients and reference function gives 17 distinct networks, not 60 independent fits. Every new uniform network-to-spline bound is below 0.001; the previous mesh-Lipschitz checks were between 0.027 and

TABLE V

COST SAVED BY REOPTIMIZATION AFTER THE INVESTMENT-PRICE CHANGE

Risk	Capital	Old action	New action	Gain lower	Gain upper
0	0.125	0.25000	0.21094	0.085570	0.085571
0	0.25	0.25000	0.16992	0.097610	0.099736
0	0.5	0.16797	0.10938	0.057079	0.057080
0	0.75	0.06836	0.05273	0.021237	0.030832
1	0.125	0.25000	0.21289	0.084553	0.084554
1	0.25	0.25000	0.17383	0.097370	0.098241
1	0.5	0.16797	0.10938	0.057067	0.057068
1	0.75	0.07031	0.05273	0.023443	0.028670

Old: conventional rule constructed at price one, held fixed at price four. New: neural-selected rule at price four. Gains compare both rules under the new price. Displayed gain endpoints are rounded outward to six decimals; action entries are descriptive.

0.031. Independent exact rational evaluation of every affine-piece endpoint verifies all 17 bounds, with two further fixtures covering zero slopes and nondyadic roots. This is an improvement in verification resolution, not a new timing experiment or an automatic improvement in the previously recorded policy-loss numbers. The audit records exact numerators and denominators, outward upper bounds, source identities and the mapping from every service to its distinct network.

13.3.0.1. *Interpretation.* The results establish a constructive NBO comparison outside exact quadratics, a residual-verified native factor implementation, and certified nonlinear reoptimization gains. They also show why a certificate is not a work-superiority result. The structural quadratic solver and the nonlinear spline baseline remain favorable comparators. General diffusion, endogenous-preference, temporal-self and game applications retain their original statements and application-specific obligations; the two completed constructions are not silently substituted for those distinct assumptions.

14. RETAINED CONSTRUCTION EVIDENCE AND EARLIER ECONOMIC COMPARISONS

This section retains the earlier exact-construction evidence and adverse comparisons. The eight-service timing contrast changes precision and caching jointly and is descriptive, not a causal precision comparison. The identified current comparisons are in Section 13.

The evidence distinguishes local decisions, full policies, and the work of constructing their continuations. The complete previous article and its technical supplement contain the original tables, including unsuccessful checks and comparisons unfavorable to NBO. The applications companion retains every economic application. We first locate the full-policy results relative to that evidence and then examine the precision-adaptive construction.

14.1. *From current decisions to full policies*

The original changed-future experiment contains 168 complete services and 840 future-specific outcomes. It evaluates current scalar actions conditional on specified future policies. Unchanged reuse certifies 72 of its 120 outcomes; check-and-refresh and fresh neural refitting each certify all 120. The 48 failed reuse checks are retained. Adaptive neural reuse costs more than fresh neural refitting in every one of the eight dimension-volume cells, with ratios between approximately 1.39 and 1.59. Cached sample-average optimization is the least-cost fully certified service in all eight cells. The signed withdrawal responses to stronger future production and valuation are valid finite-law economic results, but full-support enumeration obtains the same target more sharply and cheaply. These results support the decision certificates; they are not evidence that the neural representation is necessary for those counterfactuals.

The subsequent full-policy construction changes the comparison target, not the meaning of those earlier observations. It uses vector controls, continuous Gaussian innovations, and own-policy evaluation over 24 decision dates, with matrix inequalities covering all real states. The additive catalogue records 480 certified services and 608 failed intermediate checks among 1,088 attempts. The recursive-risk catalogue records 72 certified services under its retained execution identity. The latter cata-

logue spans two state dimensions, three risk settings, three economic regimes, two registered seeds, and two procedures. Its total recorded service work is 25.4238 seconds, with closing-ledger work reported separately. These are finite-design records, not estimates of a success probability over an unspecified population of training runs. Earlier recursive experiments that failed a domain or precision check remain preserved; later successful certificates do not erase those failures.

The full-policy comparison has no sampled-state coverage term: the coefficient inequalities hold on all states. Its policy-class approximation term is zero because the comparator is all adapted policies of finite cost. Its transfer term is zero for the directly specified discrete economy, not for an unverified approximation of the original diffusion. Evaluation, training, actor, and action-execution allowances remain distinct. A successfully verified capital policy therefore instantiates a full-policy result without asserting a certificate for a separate game or utility specification.

14.2. *A frozen comparison of continuation constructions*

The new comparison holds the economic primitives and policy tolerance fixed and varies only the construction procedure. It has two state and action coordinates, four decision dates, Gaussian covariance $I/1000$, discount factor 0.95, recursive risk parameter 0.1, and identity action and terminal cost matrices. The transition matrix is

$$A = \begin{pmatrix} 0.1 & 0.02 \\ 0 & 0.11 \end{pmatrix}, \quad B = I.$$

The anchor state cost is $Q = I$; the changed-valuation state cost is $\text{diag}(201/200, 199/200)$. Initial states satisfy $\|x_0\|^2 \leq 2$, and the requested full-policy loss is 10^{-4} . Each method solves its own future backward. In the changed economy each neural procedure starts from its own actual anchor factors, not from a competitor's solution or an optimal-policy label.

The neural comparisons retain rectangular three-by-two factors, noncommuting initial Gram matrices, nonzero evaluation radii, and perturbed actor solves. Linear warm updates and fixed-precision quadratic refresh use 40 fractional bits. Precision-adaptive refresh uses the first integer satisfying (58), subject to the registered caps

TABLE VI
COMPLETE CONSTRUCTION AT A COMMON POLICY TOLERANCE

Construction	Economy	Updates	Inverses	Bits	Gap bound	Seconds
Linear warm	Anchor	102	0	4080	1.83×10^{-14}	0.2808
Linear warm	Changed	66	0	2640	1.88×10^{-14}	0.2359
Quadratic, fixed	Anchor	6	6	240	1.53×10^{-12}	0.2224
Quadratic, fixed	Changed	3	3	120	1.01×10^{-9}	0.2250
Quadratic, adaptive	Anchor	6	3	102	1.89×10^{-12}	0.2006
Quadratic, adaptive	Changed	3	3	54	9.90×10^{-10}	0.1971
Structural	Anchor	0	0	0	0	0.0396
Structural	Changed	0	0	0	0	0.0096

Notes: All eight policies satisfy the tolerance 10^{-4} uniformly on $\|x_0\|^2 \leq 2$. Inverses count constructions for training targets only. Bits is the sum of fractional precisions across hidden updates, not total arithmetic work. Complete seconds include evaluation, training, checks, actor solves, serialization, and durable detailed output. Common startup and the final ledger are unallocated. These are deterministic construction objects, not population samples.

of 128 bits and 32 updates per fit, and builds the inverse of each fixed target at most once. The structural comparator directly solves the same quadratic economy without factor training. All methods receive the same analytic Gaussian information. The terminal transformed continuation is evaluated directly for every method.

The complete source and protocol were committed before this eight-service execution. Method order, regime order, precision limits, stopping conditions, and failure handling were fixed. A failed service would have been written and the execution terminated; no partial execution would have been reported as eight successful services. Every returned policy is checked independently by exact Bellman matrix subsolutions, in addition to its training and actor acceptance conditions. The state and action coverage is algebraic, not an interpolation claim inferred from the two-dimensional examples.

All eight services satisfy the common policy tolerance. The linear procedure uses 102 and 66 hidden updates in the anchor and changed economies. Both quadratic procedures use six and three. The adaptive procedure reduces the sum of stored fractional precisions from 240 to 102 in the anchor and from 120 to 54 in the changed

economy. Its largest precision is respectively 21 and 18 bits. For the anchor it constructs three training target inverses, rather than six; for the changed economy both quadratic procedures construct three because each trained date needs only one update. The target-inverse applications are still performed at every update.

In these two specified services the adaptive construction also has lower recorded complete time than the two inherited neural procedures. The structural procedure remains faster than every neural procedure and returns an exact zero policy-gap bound. This outcome is informative: precision and reuse improve the implemented neural construction, while the economy's structure still makes direct control particularly inexpensive. We do not convert a comparison between three neural implementations into superiority over all candidate generators, or pool these deterministic objects with historical fitting seeds to form a population test.

The service clock begins before primitive allocation and ends after the detailed record has been serialized and durably written. It includes own-policy evaluation, gates, all hidden updates, actor solves, exact policy verification, and output. Common interpreter startup and the final summary ledger are separately identified and not allocated selectively to a method. The reported inverse counts concern training targets only; Gaussian evaluation and actor solves are separately counted and included in the service clock. The fractional-precision sum is not a count of all rational bit operations. Process high-water memory is recorded as a process quantity, not as a method-specific memory frontier. No simulation-transition advantage is claimed: all eight services use analytic Gaussian evaluation and have zero simulated transitions.

The comparison supplies an executed instance of Corollary 14: a trainable continuation, constructed with explicit finite precision, yields a policy certified against full adapted controls in its stated economy. The deterministic theorem applies to every input meeting its basin, moment, and error conditions. The measured speed comparisons apply to the eight executed services. Their different scopes are part of the numerical conclusion rather than qualifications hidden from it.

15. ECONOMIC APPLICATIONS OF THE BELLMAN STRUCTURE

The motivating economic problem combines external choices with the costly adjustment of preferences. Recursive utility changes how a given policy is evaluated; preference adjustment adds an internal choice to its Hamiltonian; and strategic interaction changes whose deviations define improvement. These distinctions determine the economic interpretation of NBO. The original diffusion applications and the directly specified recursive capital economy have distinct verification accounts. The latter provides the constructive full-policy instance studied above; it is not an unverified discretization certificate for the former. The preference, temporal-self, and game applications establish the corresponding economic objects and their own verification results.

15.1. *Costly preference adjustment and portfolio choice*

Let u denote risk aversion and X financial wealth. The household chooses consumption c , portfolio share p , and preference-adjustment effort θ . The preference state is reflected on a declared interval $[\underline{u}, \bar{u}] \subset (1, \infty)$, and wealth follows the portfolio diffusion:

$$\begin{aligned} du &= \theta dt + \sigma_u dW^u + dL^{\underline{u}} - dL^{\bar{u}}, \\ dX &= \{[r + p(\mu - r)]X - c\} dt + p\sigma_X X dW^X, \quad d\langle W^u, W^X \rangle_t = \varrho dt. \end{aligned} \quad (75)$$

For a fixed consumption unit $\bar{c} > 0$, flow utility is

$$U(c, u) - \frac{1}{2}k\theta^2, \quad U(c, u) = \frac{(c/\bar{c})^{1-u}}{1-u}.$$

The objective discounts both felicity and adjustment costs at rate ρ and includes the specified wealth-exit and terminal payoff. The baseline uses $\bar{c} = 1$. Because the household controls u , the cardinal normalization across preference states is a primitive: adding a function of u changes the incentives to adjust preferences. The reflected state, positive consumption domain, and stopping data give the evaluation equation its economic and mathematical domain.

At an interior optimum with $V_X > 0$ and $V_{XX} < 0$, the three policy conditions are

$$\begin{aligned} c^* &= \bar{c}(\bar{c}V_X)^{-1/u}, & \theta^* &= V_u/k, \\ p^* &= -\frac{V_X(\mu - r)}{XV_{XX}\sigma_X^2} - \frac{\rho\sigma_u V_{uX}}{XV_{XX}\sigma_X}. \end{aligned} \quad (76)$$

Box constraints require their corresponding feasible maximizers. The cross derivative in the portfolio rule links the external risk choice to the internal preference state. Its sign is an equilibrium object to be determined by evaluation. The adjustment rule likewise relates effort to the shadow value of changing preferences. A higher adjustment cost lowers the value of every fixed policy and hence the optimized value; whenever an envelope derivative exists,

$$\partial_k V^k(t, u, X) = -\frac{1}{2} \mathbb{E}_{t,u,X}^{a^k} \int_t^\tau e^{-\rho(s-t)} \theta_s^2 ds \leq 0. \quad (77)$$

This comparative static holds with the feasible class and stopping payoff fixed.

The numerical application implements these choices jointly. The finite-action study evaluates complete policies by independent backward recursion. The continuous-action study trains the actor through its own payoff and verifies its hybrid policy with a complete action-box enclosure. Fresh training on nested grids separates the effect of resolution from the evaluation of a frozen coarse policy. A model-specific compensating-consumption calculation translates its cardinal utility bounds into an explicit transfer experiment. Application Sections [A.1.3](#), [A.1.3.4](#), and [A.1.8](#) give the full model, all recorded policies, refinement evidence, and compensation account.

15.2. Recursive utility and temporal selves

Recursive utility enters policy evaluation through the policy's own continuation utility. With $a = 1 - 1/\psi$, $b = 1 - \gamma$, and $bV > 0$, the Epstein–Zin aggregator used here is

$$f(c, V) = \frac{\rho}{a} \{c^a (bV)^{1-a/b} - bV\}. \quad (78)$$

The invariant utility interval and boundary payoff are part of the evaluation problem. Application Section [A.1.2](#) derives the homothetic policy conditions and verifies

the matching-bequest experiment, including the barriers that keep every policy utility in the aggregator's domain. This application tests recursive evaluation while maintaining the same separation between value approximation and feasible action improvement.

Time inconsistency requires a different improvement criterion. For the continuous discount kernel

$$h(s) = \beta e^{-\rho s} + (1 - \beta)e^{-(\rho+\kappa)s}, \quad 0 < \beta \leq 1, \quad \kappa > 0,$$

two critics evaluate the same future policy at the two discount rates. Writing these values as W_1, W_2 , the current self uses $V = \beta W_1 + (1 - \beta)W_2$ and chooses an instantaneous deviation by maximizing $u(a) + \mathcal{L}^a V$. Application Section A.1.4 states the complete component equations and derives the spike-deviation condition. In the homothetic consumption economy, the equilibrium consumption share is the unique root in its admissible interval, and stronger present bias raises that share. Independently optimized finite-duration deviations check the limiting equilibrium condition. Thus the actor's objective follows from the temporal equilibrium concept, while the critics retain their policy-evaluation role.

15.3. Capital investment and strategic deviations

In the dynamic Cournot application, firm i chooses investment in its own capital, with

$$dK^i = (I^i - \delta K^i) dt + \sigma_K K^i dW^i, \quad q_i = AK^i.$$

Its flow profit is $P_N(K)q_i - I^i - b_i(I^i)^2/2$, where the market-size normalization in P_N is specified for each economy. Given rivals' policies, a critic evaluates the resulting own-firm payoff and the actor changes only its own action. Consequently, the relevant economic error is the gain from a complete unilateral deviation against the frozen rival profile.

Application Sections A.1.5 and A.1.5.3 specify the boundaries, investment sets, and market normalizations, and report independent full dynamic best responses to every principal finite neural profile. The resulting profit-unit bounds include all dates and states of the stated finite game. They also cover randomized and

history-dependent unilateral deviations against those fixed Markov rivals, by finite dynamic programming. The record reports the cost of the exhaustive training oracle and any nonzero one-step equilibrium defects along with the final exploitability calculation.

These applications give the common method a precise economic interpretation. A recursive critic evaluates a utility process; preference and capital actors allocate resources and adjustment effort; a temporal actor satisfies a spike condition; and a strategic actor is assessed against unilateral best responses. Each application supplies the boundary conditions, approximation class, and error account required by its own economic question.

16. CONCLUSION

Neural Bellman Operators separate policy evaluation from feasible improvement while preserving the economic meaning of the continuation being learned. Centered action errors identify what matters for a current decision; own-policy Bellman comparisons identify what is required for a dynamic policy. The constructive results connect these objects through updated hidden weights, primitive error allocations, and explicit finite arithmetic. They permit changing futures, noncommuting warm starts, two-sided continuation errors, and inexact actor equations within the stated budget.

Precision-adaptive refresh gives a finite learning schedule whose relative Gram envelope contracts quadratically and whose stored fractional precision is allocated to the economic tolerance. A fixed target inverse is reused within a fit, but evaluation of a new future, inverse applications, actor solves, and independent policy checks remain part of the computation. The new execution demonstrates this construction against full adapted-policy comparators rather than only a current action catalogue. It also shows why construction counts and complete economic work must be reported separately: the earlier adaptive-cached implementation has smaller recorded clocks in its two exact regimes, whereas the structural procedure is faster. The factorial now separates those mechanisms and does not show a stable adaptive-precision advantage.

The controlled-diffusion formulation and the applications to recursive utility, endogenous preferences, temporal selves, and games remain part of the same research program, with their respective domains and comparison conditions. The recursively risk-sensitive capital economy supplies a complete constructive instance; it does not substitute its assumptions for those of every other application. The resulting contribution is a proved and implemented path from neural continuation construction to economic policy accuracy, accompanied by transparent evidence about when the particular construction does and does not improve measured work.

The nonlinear construction establishes the same decision-to-policy logic without quadratic closure. Its endpoint verifier certifies trained continuations on every state, and its investment experiment separates mechanical repricing from independently verified gains due to reoptimization. The structural and spline comparisons remain visible. These results extend the NBO paper through a second constructive economic realization while keeping numerical certification, learned-policy acceptance and empirical work superiority logically distinct.

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